

CardioFusion-RT: A Hybrid Multimodal Deep Learning Model Combining ResNet and FT-Transformer for ECG Interpretation and Cardiovascular Disease Prediction

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
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October 2025

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

Cardiovascular disease (CVD) is a leading cause of global mortality, yet early and accurate diagnosis remains challenging due to the complexity of cardiac signals and the limitations of existing diagnostic models. Many current systems provide a binary prediction (disease vs. no disease) and struggle to effectively integrate multi-modal data or handle significant class imbalances, hindering their clinical utility. This research presents a robust, multi-modal deep learning framework for comprehensive cardiovascular risk assessment by classifying electrocardiograms (ECGs) into Normal, Borderline and Abnormal categories along with identifying ten specific cardiac conditions. Using the large-scale MIMIC-IV ECG dataset, our proposed model introduces a novel and hybrid multi-modal model named CadioFusion-RT which integrates raw 12-lead ECG signals and structured numeric clinical features. The architecture employs a dual-branch design, combining a CNN and ResNet-based architecture for temporal feature extraction from ECG waveforms with a Feature Tokenizer Transformer (FT-Transformer) to analyze clinical numeric attributes which are extracted from CardioFeature Extraction Toolkit (CFET); a custom-developed feature extraction pipeline. To address the inherent class imbalance in clinical data, we implemented strategies including a weighted random sampler and focal loss. Trained on the comprehensive MIMIC-IV ECG dataset, the CadioFusion-RT model achieves exceptional performance, with AUROC values exceeding 0.95 across categories including normal, borderline and abnormal ECGs as well as specific conditions like myocardial infarction, tachycardia, bradycardia and arrhythmias. By addressing limitations of single-modal approaches such as diagnostic gaps and data imbalance—the framework delivers robust, scalable solutions for early intervention, personalized treatment and real-time deployment in clinical and remote settings. This work not only outperforms state-of-the-art models in accuracy (up to 99.7% for certain tasks) and efficiency but also bridges the gap between AI-driven diagnostics and ethical, interpretable healthcare, paving the way for transformative CVD management.

Keywords: Cardiovascular Disease, Deep Learning, Multi-Modal, Electrocardiogram, Convolutional Neural Network, Feature Tokenizer-Transformer, Residual Network, Feature Database, Densely Connected Convolutional Network, MIMIC-IV

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Nomenclature

The next list describes several symbols & abbreviation that will be later used within the body of the document

AUPRC Area Under the Precision-Recall Curve

AUROC Area Under The Receiver Operating Characteristics Curve

Bi – LSTM Bidirectional Long Short-Term Memory

CFET CardioFeature Extraction Toolkit

CNN Convolutional Neural Network

CVD Cardiovascular Disease

ECG Electrocardiogram

HR Heart Rate

LBBB Left Bundle Branch Block

LSTM Long Short-Term Memory

LVH Left Ventricular Hypertrophy

MI Myocardial Infarction

MIMIC – IV Medical Information Mart for Intensive Care, version IV

MLP Multilayer Perceptron

PAC Premature Atrial Contractions

PVC premature Ventricular Contraction

RBBB Right Bundle Branch Block

ROC Receiver Operating Characteristic

SD Standard Deviation

SHAP SHapley Additive exPlanations

Chapter 1

Introduction

1.1 Background

Cardiovascular disease refers to a range of disorders affecting the heart and blood vessels. Cardiovascular disease is a leading cause of death worldwide. According to the World Health Organization, CVD causes 31% of deaths annually in the world [13]. Moreover, many people remain undiagnosed for a long time unless any life-threatening event happens. Also, those who survive heart attack or stroke often suffer from long-term life-threatening complications resulting in a reduced lifetime. CVD also economically impacts people, with billions spent annually on treatment, hospitalizations and long-term care. Despite significant advancements in medical science, CVD remains a major concern due to its complex nature and the challenges in early detection. Several models have been developed for early diagnosis of cardiovascular disease. However, the models give a prediction of whether the person is suffering from any cardiovascular disease or not. The models do not clearly classify cardiovascular diseases into different types. We are developing a technique to detect different types of CVD specially Arrhythmia, Myocardial Infarction (MI), Left Bundle Branch Block (LBBB), Right Bundle Branch Block (RBBB), Bradycardia, Tachycardia, AV (atrioventricular) Block, Left Ventricular Hypertrophy (LVH), Premature Atrial Contractions (PAC) and Premature Ventricular Contraction (PVC).

1.2 Rational of the Study or Motivation

Current ECG classification systems have serious problems that prevent healthcare providers from using them effectively. Most existing systems only classify results as normal or abnormal. This approach fails to address the clinical reality where borderline cases requiring nuanced interpretation are common. This creates a big gap between what the model or system says and what doctors actually need for making decisions. Single-modal models that use only raw ECG signals miss important clinical information. This leads to lower diagnostic accuracy and makes doctors less willing to trust these systems. Medical datasets often have class imbalance problems which makes models fail to detect diseases properly which can be life-threatening. Many deep learning models work like black boxes which is a major problem for clinical use. Healthcare providers need systems that can explain how they make decisions and show their reasoning process. Without this trans-

parency, doctors cannot trust or adopt these technologies. This study addresses these significant gaps by introducing a comprehensive multimodal framework. The framework combines raw ECG signals with clinical numeric features using a new hybrid ResNet-FT-Transformer architecture. Instead of just normal versus abnormal, it uses a three-class system: normal, borderline and abnormal. It can also classify specific diseases which directly matches what clinicians need for interpretation. This research goes beyond just technical improvements. It could lead to better cardiovascular care through earlier detection, reduced healthcare disparities and improved diagnostic capabilities across different healthcare settings. This work takes an important step toward creating AI-driven cardiovascular diagnostics that doctors can actually use in practice to help manage the global burden of cardiovascular disease.

1.3 Problem Statement

Cardiovascular diseases cause about 17.3 million deaths worldwide every year [15]. Despite medical advances, quickly identifying and accurately assessing CVD risks remains a major challenge. Using single methods like ECG analysis with echocardiography and clinical biomarkers creates diagnostic gaps and errors. Traditional machine learning models rely on handcrafted feature extraction methods. This limits their ability to identify complex relationships between physiological patterns when predicting CVD. Earlier studies have used CNNs, LSTMs and ensemble models for deep learning in cardiovascular risk prediction [13] [30]. Current medical decision-making approaches have serious problems. They depend on single data types and work with biased data. These systems also cannot explain how they make decisions. Multi-modal frameworks that combine ECG signals with clinical demographics and imaging data show promise but they face challenges with feature fusion, real-time adaptability and clinical interpretability. Existing models have trouble generalizing across different patient groups. They struggle with heterogeneous datasets and imbalanced data distribution patterns [16]. This research presents a deep learning-based multi-modal analysis framework to fill these gaps. The framework uses ECG signals and clinical biomarkers to improve cardiovascular disease classification. The study develops advanced feature fusion techniques, transformers, attention mechanisms and optimization methods with learning rate schedulers to boost predictive accuracy and interpretability. The proposed model uses cutting-edge deep learning architectures to create a more reliable, explainable and scalable solution for cardiovascular disease prediction. This contributes to early intervention and personalized healthcare.

1.4 Objective

This research aims to develop a comprehensive multimodal deep learning framework that integrates raw 12-lead ECG signals with clinical biomarkers to achieve accurate cardiovascular risk assessment and precise multi-class disease classification. The primary objectives include:

Create an advanced deep learning system that performs three-class ECG classification (Normal, Borderline, Abnormal) while simultaneously detecting specific cardio-

vascular conditions including arrhythmia, myocardial infarction, tachycardia, bradycardia, AV-block, LBBB, RBBB, PAC, PVC and Left ventricular hypertrophy.

Technical Objectives:

1. Develop a dual-input architecture combining ResNet-based signal processing with FT-Transformer for clinical features integration
2. Implement advanced feature fusion techniques with attention mechanisms to optimize predictive accuracy and interpretability
3. Address class imbalance challenges through weighted sampling and focal loss strategies
4. Achieve superior performance metrics with AUROC values exceeding 0.95 across disease categories

Clinical Objectives:

1. Enable early intervention through precise risk stratification and borderline case identification
2. Ensure scalability for deployment in diverse healthcare environments from ICUs to remote monitoring systems
3. Bridge the gap between AI-driven diagnostics and clinical practice through interpretable, actionable predictions that support personalized healthcare and reduce diagnostic errors in cardiovascular disease management

1.5 Methodology in Brief

This study employs a quantitative, data-driven methodology centered on deep learning to analyze cardiovascular data. The research was conducted through the following key stages:

1. **Data Acquisition and Preprocessing:** Raw ECG signals in WFDB format were converted to NumPy arrays for efficient processing, while clinical features were extracted using CardioFeature Extraction Toolkit (CFET). Preprocessing included noise reduction through Butterworth filtering, standardization, missing value imputation and data augmentation techniques (Gaussian noise, temporal shifting, amplitude scaling) to enhance model robustness.
2. **Multimodal Architecture Development:** A dual-input deep learning framework was designed combining a ResNet-based 1D CNN branch for ECG signal processing with an FT-Transformer branch for clinical features. The ResNet component utilizes residual connections, batch normalization and global average pooling to capture temporal-spatial patterns, while the FT-Transformer employs self-attention mechanisms to model inter-feature relationships in numeric data.

3. **Training Strategy:** The model training addresses class imbalance through weighted random sampling and focal loss implementation. AdamW optimizer with OneCycleLR scheduling ensures stable convergence, while mixed precision training optimizes computational efficiency. Early stopping based on validation F1-score prevents overfitting.
4. **Evaluation Framework:** Performance assessment utilizes comprehensive metrics including accuracy, precision, recall, F1-score, AUROC, AUPRC and specificity. Threshold optimization maximizes F1-score on validation data, ensuring clinical relevance.
5. **Implementation and Validation:** The framework undergoes rigorous testing using stratified train-validation-test splits, with final evaluation on completely withheld test data to ensure unbiased performance assessment and real-world applicability.

1.6 Scopes and Challenges

Research Scope: This study focuses specifically on automated ECG classification using the MIMIC-IV dataset for three-class categorization (Normal, Borderline, Abnormal) and ten disease-specific conditions. The scope encompasses multimodal integration of 12-lead ECG signals with extracted clinical features, addressing cardiovascular risk assessment in ICU and clinical environments. The framework targets deployment scenarios including real-time monitoring systems, clinical decision support tools and telemedicine applications.

Technical Scope:

1. Integration of ResNet and FT-Transformer architectures for multimodal learning
2. Scalable deployment across various computational environments
3. Support for both batch processing and real-time inference capabilities

Data-Related Challenges:

1. Significant disparity between normal cases and rare cardiovascular conditions requiring specialized sampling and loss function strategies
2. Variability in ECG signal quality, patient demographics and recording conditions affecting model generalization
3. Incomplete clinical features and corrupted ECG signals necessitating robust imputation methods

Technical Challenges:

1. High memory and processing requirements for training multimodal deep learning models on large-scale datasets

2. Optimal fusion of temporal ECG signals with structured clinical data without information loss
3. Balancing deep learning performance with clinical explainability requirements for healthcare adoption

Resource Constraints:

1. Need for high-performance GPUs and substantial memory for model training and deployment
2. Extended computational time required for convergence on large multimodal datasets, limiting rapid prototyping and iteration capabilities

These challenges collectively define the boundaries within which the research operates, acknowledging limitations while establishing clear pathways for future enhancement and clinical translation.

Chapter 2

Literature Review

2.1 Preliminaries

ECG also known as Electrocardiogram is a test that records the electrical activity of the heart over time. During an ECG, small adhesive electrodes are placed on the chest, arms and legs. Lead wires connect these electrodes to the ECG machine which measures and prints the electrical activity of the heart. No electricity is sent into the body. The natural electrical signals coordinate each contraction to keep blood moving properly. An ECG captures these signals to show how fast the heart is beating whether the rhythm is regular or irregular. Changes in the ECG can indicate a wide range of heart conditions [2]. A sample ECG is shown in Fig. 2.1.

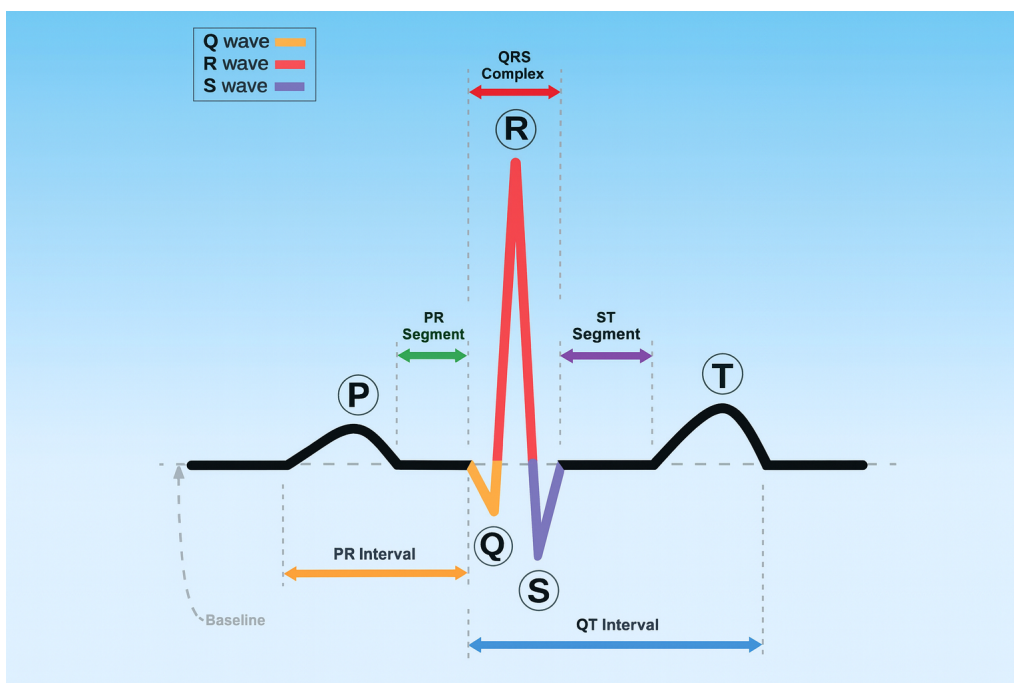


Figure 2.1: Electrocardiogram

Convolutional Neural Network (CNN) is a deep learning model designed to interpret grid-like data, most commonly images. CNNs excel at identifying visual patterns, finding smaller objects and grouping images by learning layered spatial

features.

Deep Neural Network (DNN) is an advanced neural network. It has multiple hidden layers due to which it can learn complex patterns from data. It works by passing information through multiple layers of connected nodes, where each layer pulls out meaningful features step by step. The deeper the network, the better it can capture complex relationships, making DNNs powerful in recognizing and classifying images, processing speech and predicting diseases.

SHapley Additive exPlanations (SHAP) is a method for predicting interpretations for machine learning models. It is derived from game theory and allows explanation of individual prediction using the output of a model from the input feature. The SHAP values mainly indicate the relevant feature that contributes to the model prediction.

FT-Transformer (Feature Tokenizer + Transformer) is an adaptation of the Transformer architecture designed specifically for tabular data. Rather than treating rows as sequences of time steps, FT-Transformer tokenizes each feature (both numerical and categorical) into a learned embedding via a feature tokenizer; categorical features receive embeddings and numerical features are linearly projected (optionally with encoding schemes) into token vectors. These feature tokens — together with a special [CLS] token — are processed by a stack of Transformer encoder layers (multi-head self-attention + feed-forward blocks), so the model learns pairwise and higher-order interactions between features via attention. The final representation of the [CLS] token is used for classification or regression. An architecture of FT-Transformer is shown in Fig. 2.2.

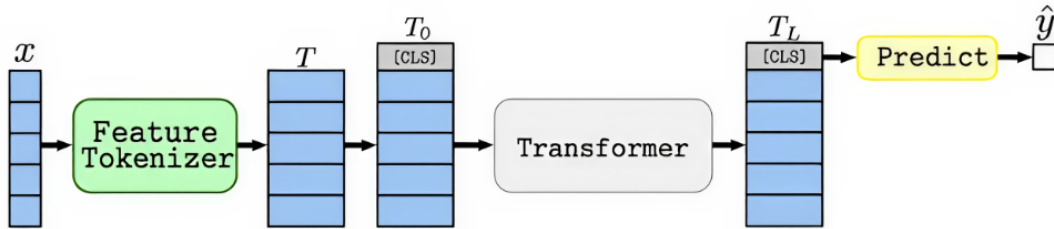


Figure 2.2: FT -Transformer

Residual Networks (ResNet) are deep convolutional neural networks that address the problem of vanishing gradients in very deep architectures by introducing skip connections. These connections allow information to bypass one or more layers, enabling the training of networks with hundreds of layers without degradation in performance. ResNets have been widely used in medical signal and image analysis due to their ability to extract hierarchical representations of data. In ECG classification, ResNet can efficiently learn both low-level waveform features (such as QRS complexes) and high-level abstractions (rhythm patterns) which lead to improved accuracy for abnormality detection.

Densely Connected Network (DenseNet) is an advanced form of convolutional

neural networks (CNNs). In this architecture, every layer is directly linked to all previous layers in a feed-forward manner. While ResNet uses skip connections, DenseNet allows each layer to reuse the feature maps of previous layers. This approach minimizes redundancy and supports feature reuse. This results in fewer parameters and improved gradient flow during training. When Densenet is applied to biomedical signals such as ECG and PCG, it can recognize slight variations across leads or signal cycles. These capabilities make it effective in detecting small but clinically important abnormalities [9].

Bi-LSTM or Bidirectional Long Short-Term Memory extends the basic LSTM architecture designed for sequential data analysis. Traditional LSTMs read data in a single direction. But Bi-LSTM processes sequences in both forward and backward directions. This dual-directional learning allows the model to capture past and future dependencies in the data, which is particularly valuable in ECG signal analysis where both preceding and succeeding waveform patterns provide diagnostic clues. Bi-LSTM has shown strong performance in arrhythmia detection and heartbeat classification tasks, as it preserves long-term dependencies while handling variable-length sequences efficiently [8].

Feature Database (FeatureDB) is a structured repository of extracted features derived from raw biomedical signals such as ECG or PCG. Instead of storing the entire time-series signal, FeatureDB contains relevant summary descriptors such as heart rate (HR), QRS duration, QT/QTc interval, PR interval and axis measurements along with their corresponding diagnostic labels [31].

Arrhythmia is a broad term for any disorder of the heart’s rhythm, where the heart beats too fast, too slow, or irregularly. It’s caused by a disruption in the heart’s electrical signaling system. While some arrhythmias are harmless, others can be life-threatening, increasing the risk of stroke or heart failure. On an ECG, it manifests as an irregular pattern of QRS complexes [17].

Bradycardia is a specific type of arrhythmia characterized by an abnormally slow heart rate, typically defined as fewer than 60 beats per minute in an adult. This condition can lead to insufficient blood flow to the brain and other organs, causing symptoms like fatigue, dizziness and fainting. On an ECG, it is identified by a normal sinus rhythm but with an extended interval between heartbeats (a low heart rate) [4].

Tachycardia is the opposite of bradycardia, defined as a heart rate that exceeds the normal resting rate, typically over 100 beats per minute. This rapid beating reduces the heart’s efficiency, impairing blood flow to the rest of the body and increasing the work of the heart muscle. ECGs reveal a heart rate significantly above normal, which can originate from either the atria or the ventricles [4].

Myocardial Infarction (MI) also commonly known as a heart attack, Myocardial Infarction is a life-threatening condition that occurs when blood flow to a part of the heart muscle is blocked, causing tissue death. An ECG is a critical tool for diagnosing MI, which often presents with characteristic changes like ST-segment elevation

or depression, and pathological Q-waves, indicating irreversible muscle damage [21].

Atrioventricular (AV) block is a delay or interruption in the electrical signal that travels from the atria to the ventricles. It can be mild and cause few or no symptoms, or it can be complete and stop the atria and ventricles from communicating which is a medical emergency. On an ECG, AV block often appears as a prolonged PR interval.

Left Bundle Branch Block (LBBB) is a defect in the heart's electrical conduction system where the signal is delayed or blocked as it travels down the left bundle branch to the left ventricle. This forces the signal to take a slower, less efficient route, causing the left ventricle to contract later than the right. The hallmark ECG finding is a widened QRS complex (typically ≥ 120 ms) with a characteristic shape in the chest leads [3].

Right Bundle Branch Block (RBBB) involves a block or delay in the right bundle branch, affecting the right ventricle. The electrical impulse must travel from the left ventricle to activate the right, causing a delay in right ventricular contraction. This also produces a widened QRS complex on the ECG but with a different pattern (e.g., an "M" shape in lead V1) that distinguishes it from LBBB [3].

Left Ventricular Hypertrophy (LVH) is the thickening and enlargement of the muscle walls of the heart's main pumping chamber, the left ventricle. It often develops in response to chronic high blood pressure or heart valve problems, which force the heart to work harder. While an adaptive response, it increases the risk of arrhythmia and heart failure. ECG criteria for LVH include increased QRS voltage and amplitude.

Premature Atrial Contraction (PAC) is a common and usually benign event where an extra, premature heartbeat originates from the atria, disrupting the normal rhythm. It feels like a "skipped" or "extra" beat. On an ECG, a PAC is identified by a premature and often abnormally shaped P-wave, followed by a QRS complex that may or may not be normal.

Premature Ventricular Contraction (PVC) is an extra heartbeat that originates from the ventricles rather than the atria. These are also common and often harmless in healthy individuals but can indicate underlying heart disease if they are frequent. An ECG shows a premature and bizarrely shaped QRS complex that is not preceded by a P-wave, followed by a compensatory pause.

2.2 Review of Existing Research

The study of Li et al. (2020) [6] mainly focuses on the significance of detecting CVD in the early stage using CraftNet, a deep learning ensemble model as the existing models struggle with dissimilar arrangement of classes and categories as well as with the new, unseen data which are outside the training data while analyzing ECG signals. This study used the MIT-BIH dataset containing 48 30-minute-long ECG records of 47 patients which is at 360 Hz. The heartbeat classes are expressed in 4 categories which include N(normal beat), S(supraventricular ectopic beat), V(ventricular ectopic beat), F(fusion beat) and Unknown beat. The dataset contains 90% N-type heartbeats comprising data imbalance. To alleviate the problem of data imbalance and for increasing accuracy rate CraftNet merged both handcrafted features such as R-R intervals, wavelet transformations, higher order statistics and morphological data with DDAG structure and novel P-S loss(pseudo-SVM loss). The model had shown average sensitivity of 89.25% and specificity of 95.79% which is better than the established models establishing robustness against imbalanced data. Despite showing remarkable results the main drawback of this model is it is not completely automated as it uses handcrafted features which is hindering its adaptability to new datasets and not all datasets are prone towards manual feature extraction. Moreover, it has a complex training process for using mixed loss functions requiring more time in careful fine tuning though it is improving the performance. Not only this there may be high possibility of not handling the larger or multi-lead ECG datasets by the model which will limit it from adapting to real-world clinical applications. For improvement of this model, these issues need to be addressed i.e. leading it towards end-to-end adaptability and optimizing the architecture.

The paper of Ordás et al. (2023) [10] emphasizes on predicting CVD from datasets that contain a low number of features. The dataset used in this paper is a combination of datasets of Cleveland, Hungarian, Switzerland, Long Beach, Stalag. A sparse autoencoder is used for increasing features of the data. Then 2 different classifiers MLP and 2D CNN are used to predict whether the person has CVD or not. The MLP classifier with SAE gains 89.543% accuracy which is 3.78% higher than MLP without sparse autoencoder. When 2D CNN is used as a classifier with SAE, it achieves 90.09% accuracy. These 2 results are compared to other ML algorithms used to predict CVD such as Random forest, KNN, Decision tree etc and the combination of sparse autoencoder and 2D CNN proves to be the most accurate. Even after combining multiple datasets, it has less than 1000 samples. So it can impact the generalizability of the model. Moreover, it is difficult to interpret how input features influence prediction. So clinical adoption can be hindered because of lack of interpretability.

Hossain et al. (2023) [12] introduced a hybrid deep learning framework through the integration of CNN and LSTM networks for capturing spatial and temporal correlations from the data which would lead to CVD detection. Advanced feature engineering and Explainable AI (XAI) are incorporated into this model. Critical factors which would influence the prediction rate are identified using SHAP (SHapley

Additive exPlanations). A dataset from Kaggle with 70,000 samples from patients was used where 49.97% were heart disease samples and 50.03% were normal samples. The model achieved an accuracy rate of 74.15% with feature engineering and 73.89% without feature engineering, more than the rate shown by RF, XGBoost and SVM. The study worked on the previous gaps of the former studies by creating a hybrid architecture where CNN and LSTM were used for extracting features and capturing sequential patterns respectively with the help of feature engineering. The model is mainly dependent on a single dataset which is limiting its potential of generalizability over large and diverse datasets. SHAP has been used in the study for in-depth analysis of features but its real application is being missed in case of clinical decision making. Though the accuracy rate is better than some of the previous algorithms, it is not giving the optimal accuracy rate in recent times leaving a lot of scope for improvement. The model needs to be incorporated with some advanced deep learning models as well as semi-supervised learning, it can also be expanded to diverse datasets which would eventually optimize the computational performance improving versatility and scalability of the model.

Lee et al. (2023) [13] developed a model which uses multi-modal data for predicting Cardiovascular Disease. Dataset from the Samsung Medical Center (SMC) is used for development and internal validation while UK biobank dataset is used for external validation. 11 different approaches are used and compared to find out the best model with the highest accuracy. DenseNet-169 is trained on fundus photographs (FP) and has an AUROC of 0.548. Logistic regression is trained on only clinical risk factors which attains the AUROC of 0.854. DNN is trained using CRF and gained an AUROC of 0.849. DenseNet-169 and logistic regression is trained on FP and clinical risk factors (CRF) respectively which attains 0.792 AUROC. DenseNet-169 and DNN are trained on fundus photographs and clinical risk factors which obtains 0.872 AUROC. 10-year ASCVD risk of PCE model was trained on CRF and smoking history which was not present in SMC dataset. It obtains 0.677 AUROC. DenseNet-169 is trained on FP and DNN is trained on FP which attains 0.882 AUROC. DenseNet-169 combined with DNN integrates uncertainty quantification and achieves AUROC of 0.905. Invasive CRF gives better AUROC than non-invasive CRF. However, the study has clear limitations, such as the lack of diversity in datasets, which limit the generalizability of the findings across different populations. Deep learning models consist of multiple hidden layers, due to which the result lacks clinical interpretability and trust. The shortcomings include the lack of consideration for other potentially valuable imaging modalities or risk factors that could further enhance predictions. Future work should focus on incorporating more diverse datasets collected from different regions, developing interpretable models to enhance clinical adoption, and validating these models in real-world clinical settings to assess their feasibility and cost-effectiveness in routine practice.

The study of Alsekait et al. (2024) [15] aims at diagnosing heart disease by introducing Heart-Net, a multi-modal deep learning framework which would increase the accuracy rate and consider the relevant factors by integrating the MRI and ECG signals to properly utilize the effect of each method. A 3D U-Net and TCGNN (Temporal Convolutional Graph Network) are used for extracting features from MRI signals and ECG data respectively. The model achieved accuracy rates of 92.56%, 93.45%

and 91.89% for the respective 3 datasets HNET-DSI, HNET-DSII and HNET-DSIII outperforming existing frameworks such as fuzzy logic, genetic algorithms etc. which were limited by many constraints such as manual feature engineering. Though the achievements of the model are significant, it has some limitations in adaptability across new datasets. Despite using techniques like ADASYN to reduce imbalanced data, the risk of overfitting prevails due to the complex algorithms of the model. It is quite difficult for the medical professionals to go through the outcomes of the model due to a lack of interpretability. These limitations can be solved by using wearable devices' real time data as well as incorporating transformer-based models that can extract complicated patterns increasing scalability as well as the accuracy rate for detecting CVD in the early stage and other risks of the model. Though the proposed model has shown remarkable outputs, further optimization is required for increasing its robustness and efficiency.

Honi et al. (2024) [18] proposed a one-dimensional convolutional neural network approach which uses UCI repository heart disease dataset for predicting CVD at an early stage. After pre-processing the dataset, Chi-square ranking method is utilized to assess the significance of various features in predicting CVD. The model uses convolutional layers for extracting significant features. Then extracted features are fed into a batch normalization layer. For reducing the chance of overfitting, dropout layers are used. The extracted features are flattened and passed through dense layers, with a ReLU activation in the first layer to capture complex patterns and a sigmoid activation in the final layer to output the probability of CVD presence. To evaluate the efficiency of the model, 3 approaches are used. A train-test split, 5-Fold cross-validation and 10-Fold cross-validation. Using the train-test split method, the model gained 99.95% accuracy, 99.93% precision, 99.96% recall and 99.97% F1-score. 5-Fold cross-validation provides less accuracy compared to 10-Fold cross-validation. 5-Fold cross-validation provides 98.53% accuracy rate, 99% precision, 99.92% recall, 99.91% F1-score while 10-Fold cross-validation achieves 99.12% accuracy rate, 99.94% precision, 99.93% recall and 99.98% F1-measurement. The study used the UCI Heart Disease dataset with only 1025 samples, so the model may not have the capacity to generalize to larger, real-world datasets with more complex and diverse patterns. Selecting hyperparameters causes uncertainty. Biased datasets can harm the model's performance. Using Electronic Health Record (EHR) data raises concerns about data completeness and accuracy, which can impact the model's reliability. To improve generalization, datasets from multiple sources can be incorporated in future. Also, the impact of the model can be evaluated by testing the model with live patient data.

Liu et al. (2024) [19] has shown a proposal of integrating multimodal data with deep learning in diagnosing heart failure. It mainly works by combining multiple factors such as ECG signals, X-rays of the chest and structured text data as it can outperform the single modality models in terms of accuracy rate. A dataset was collected from a hospital which included 440 cases where 325 belonged to heart failure patients and 115 were of mild CVD. The mentioned multimodal model works mainly by integrating features leveraging both feature level and score level fusion. For ECG signal processing, chest X-ray analysis and structured text data, it uses 1D-CNN, DenseNet121 and MLP individually. This model has a high accuracy rate

of 99.63% when it was tested with data augmentation and it has also shown robustness against noise which eventually outperformed the former single as well as the dual-modal models, depicting the significance of introducing data fusion in the model. Despite significant outcomes, the model limits some factors like less adaptability to diverse datasets, reducing reliability on the performance due to the absence of external validation and overfitting can be introduced due to extensive use of data augmentation. It also has no clear analysis on handling corrupted data or noises. Future improvement can be done by exploring hybrid fusion techniques that will improve the performance as well as cross data validation needs to be implemented. Adversarial training and expanding the dataset using multi-center collaborations can help to reduce overfitting. GANs (Generative Adversarial Networks) can be used for handling the biases. Lastly, KNN-imputations or Bayesian models can be used to work with corrupted or missing data.

A new multi-modal framework has been introduced in this study Revathi et al. (2024) [22] which works through combining demographic data with ECG signals for predicting heart disease with an improved accuracy rate. The PTB-XL dataset has been used which contained 21,000 12-lead ECG samples from 18,869 patients. The model is mainly based on working with different signal preprocessing techniques which include IIR (Infinite Impulse Response) filtering and feature engineering integrating with FFT which is needed for extracting pivotal frequency dependent features. Using the MLP (Multi-layer Perceptron) model in the extracted refined features, it showed an optimal performance with precision of 79%, recall of 73% and F1 score of 76% in the real-world clinical scenarios. Traditionally, models like CNN, RNN or any other ensemble model as well as hybrid models like CNN-LSTM have shown high accuracies but they were limited in working with imbalanced data and there were other research gaps which were filled by the mentioned model through integration of refined feature extraction and selection, robust feature engineering and multi-modal techniques. However, the model fails in capturing the complicated patterns fully in the ECG signals as it is more dependent on feature extraction and selection strategies. Moreover, it is not ideal for handling the real-time ECG monitoring system. Future improvements can be made by introducing the model to larger datasets as well as incorporating real-time data from wearable devices for dynamic diagnosis and also integrating MLP with CNN and RNN which would enhance both demographic and ECG data while detecting heart diseases. These future working probabilities can advance the procedure of CVD diagnosis.

The paper Talaat (2024) [23] proposes a self-attention based transformer model called CardioSentiNet for predicting cardiovascular disease. The dataset used to train the model is generated from a selected group. After preprocessing the dataset, the data is passed to a self-attention based transformation module. Then the transformed dataset is passed to the CNN layer. Then the parameters are fine-tuned. The Training and Optimization step fine-tunes the model's parameters using a training dataset to gain optimal accuracy. Interpretability techniques are added to maintain transparency. Finally, the validation and testing phases assess the model's generalizability and performance on unseen data. Multi-layer neural network with ReLU activation function is used as model architecture. The proposed model is evaluated using R^2 -value and then compared with other models through this evaluation met-

ric. The reported R^2 -value for the proposed model is 0.994 while other models like CardioNet-deep, CNN, RNN etc have less R^2 -value value than the proposed model. The dataset does not represent the diverse population which limits the ability to generalize the result. High R^2 -value raises concerns about overfitting. The study depends on a single dataset without any external validation. In future, broader demographic representation can be incorporated into the dataset To improve dataset, data augmentation method can be used. To guarantee broader applicability, stratified validation can be executed.

Xia et al. (2024) [24] proposed ICVD-ACOEDL (Intelligent Cardiovascular Disease Diagnosis) model which uses multiple modalities in diagnosing heart diseases. For feature selection, classification and hyperparameter tuning it uses ACO (Ant Colony Optimization), DLENN (Deep Learning Enhanced Neural Network) and Bayesian Optimization correspondingly. A dataset from Kaggle was used which contained 1190 samples including 629 disease affected and 561 normal samples. The proposed model surpassed the earlier methods like- ANN, SVM, bagging and Naive Bayes with an accuracy rate of 99.71%. ICVD-ACOEDL model has introduced both automated optimization process and feature selection which enhanced the efficiency and scalability of the model in the real-world data. This model will face some challenges like losing important details using ACO though it assists in improving feature selection and using a single dataset will reduce the diversity of the model. Scaling in the case of larger datasets is tough and challenging due to the requirement of detailed hyperparameter tuning in the model. Improvements can be made by integrating methods for identifying outliers and data clustering in order to increase the effectiveness of the classification of the model, exploring transformer-based architectures and expanding datasets for enhanced robustness. By overcoming these issues, the ICVD-ACOEDL model has the potential to transform CVD diagnostics, providing a scalable and highly accurate approach.

The sole target of this study Zhu et al. (2025) [32] is to focus on the multi-modal approach by integrating ECG and PCG signals to detect cardiovascular disease rather than using single-modal approaches. This study used the 2016 PhysioNet/CinC Challenge dataset which contained 6 data subsets from training-a to training-f where training-a contained 409 records. These records contained the ECG and PCG signals of both normal people and patients. The DDR-Net model is being introduced in this study which mainly consists of a dual scale feature aggregation module, a deep residual block module and a Bi-LSTM layer. At first, the DDR-Net- Dual Scale Deep Residual Network model had been used for extracting different features from the signals. Dual scale feature aggregation module mainly detects all kinds of patterns small or large and combines them from the signals using Bi-LSTM layer so that all the information can be extracted. Then for eliminating irrelevant and noisy features SVM based RFE (Recursive Feature Elimination) is used and then through cross validation overfitting is reduced. Lastly, after selecting the features through the SVM classifier decision is made whether a heart signal is a normal or abnormal condition. The model showed 91.6% accuracy rate and an AUC of 0.962 which is quite higher accuracy rate than other single-modal and existing multi-modal models. It also showed 92% sensitivity value, 91.1% specificity value and 0.915 F-score combining both ECG and PCG signals. But the failings of this model are that it is

mainly specified to a particular dataset and also there is an absence of cross validation which would limit its adaptability to real-world factors. Using SVM-REFCV would cause computational overhead making the model unfitting for larger datasets. Though the model improves the accuracy rate using Bi-LSTM, increasing the complexity and reducing the clarity of the predictions to the professionals. Further improvements can be made by integrating the medical history, demographics and other factors of a patient with the ECG and PCG signals which will increase the accuracy rate of the model.

The paper Nowroozilarki et al. (2024) [20] presents a data-efficient framework for identifying cardiac arrhythmias. The authors address the high cost of creating large labeled medical datasets by employing a supervised contrastive learning approach. Their model uses a ResNet50 encoder to create a “morphology-aware” embedding space where ECG signals with the same abnormality (like Atrial Fibrillation, Sinus Bradycardia, or Sinus Tachycardia) are clustered together. This pretraining phase requires only 10% of the labeled data but enables the model to learn distinct physiological features without relying on data augmentation that could distort the signals. In their findings, this method significantly outperforms fully supervised models, achieving a macro AUROC of 0.96 and a balanced accuracy of 0.91, compared to the 0.89 AUROC and 0.86 accuracy of a fully supervised ResNet50 baseline. Key limitations include the use of machine-generated labels and analysis of only a single ECG lead. The authors suggest future work should involve incorporating a wider range of abnormalities, using NLP to process cardiologist reports for better labels, and analyzing all 12 ECG leads for a more comprehensive diagnosis.

The study by Nayyab et al. (2025) [28] presents a deep learning-based framework for accurate multi-class classification of cardiovascular diseases (CVDs) using electrocardiogram (ECG) signals. The authors utilized the large-scale PTB-XL database, filtering and preprocessing 17,232 ECG records with a Butterworth bandpass filter (1–45 Hz) and Discrete Wavelet Transform (DWT, db8) to remove baseline drift and noise while preserving key waveform characteristics. Morphological features based on P-QRS-T intervals and amplitudes were extracted using the Pan-Tompkins algorithm, resulting in a 99-feature dataset. To mitigate class imbalance among five diagnostic categories—Normal (NORM), Myocardial Infarction (MI), Hypertrophy (HYP), Conduction Disturbance (CD), and ST-T Changes (STTC)—the SMOTE-NC technique was applied. Two models, a Convolutional Neural Network (CNN) and a Deep Neural Network (DNN), were trained with five-fold cross-validation, achieving mean accuracies of $81\% \pm 0.03$ and $84\% \pm 0.01$, respectively. The DNN demonstrated superior sensitivity (83.7%) and specificity (95.9%), with particularly high accuracy in detecting Hypertrophy (98%). The study concludes that combining advanced signal preprocessing with morphological feature extraction and deep learning yields a robust and computationally efficient approach for multi-class ECG disease detection, supporting its potential integration into real-time clinical diagnostics.

The paper by Ruiz-Barroso et al. (2025) [29] proposes a novel deep learning system, FADE, designed for forecasting normal electrocardiogram (ECG) signals and detecting anomalies in a self-supervised manner—eliminating the need for large labeled

datasets. The model combines SlowFast and U-Net architectures to forecast future ECG segments, trained using a morphology-aware Split-MSE loss function that separately penalizes errors in R-peaks and inter-peak intervals. The approach was trained on the MIT-BIH Normal Sinus Rhythm (NSR) database and adapted to the MIT-BIH Arrhythmia dataset through a domain adaptation process, enabling the model to generalize across different sensors and subjects. Anomalies were detected using a custom Normalized Mean Absolute Error (NMAE) distance metric comparing predicted versus real signals. Experimental results showed that the domain-adapted FADE model achieved a global accuracy of 85.45%, with 83.95% for anomalies and 86.95% for normal signals—outperforming transformer-based baselines by about 17%. The study demonstrated robustness across varying signal lengths and strong capability in identifying both beat- and rhythm-related abnormalities. However, the authors acknowledged limitations, including the model’s need for several seconds of normal data before forecasting and its inability to classify specific anomaly types, only distinguishing normal from abnormal patterns. Overall, FADE represents a computationally efficient, label-free ECG anomaly detection framework with potential applications in wearable and real-time cardiac monitoring systems.

The study by Abubakar et al. (2025) [25] presents a novel deep learning framework called ECGNet for high-precision classification of cardiovascular diseases using ECG image data. A dataset of 1,937 ECG images was collected from various hospitals in Pakistan using EDAN SERIES-3 ECG devices, including five classes: Normal Sinus Rhythm, Arrhythmic Heartbeat, Myocardial Infarction (MI), Previous History of MI (PMI), and COVID-19-related cardiac abnormalities. The data were preprocessed through image resizing (244×244), gamma correction, hue and brightness adjustments, and Gaussian filtering, followed by manual annotation using the LabelIMG tool. The proposed ECGNet model incorporates an embedding layer for feature extraction and two dense layers (256 and 128 neurons) activated by an Advanced Exponential Linear Unit (A-ELU) to address vanishing gradient issues, along with a Modified SparseMax classifier for improved probability mapping. The model was compared with three other architectures—TabNet, GRU, and LSTM—trained under identical hyperparameters. Results showed that ECGNet achieved 99% accuracy, outperforming TabNet (98%), GRU (96%), and LSTM (95%), with nearly perfect precision and recall for all classes except COVID-19 (recall 0.92). The study concludes that ECGNet offers superior accuracy and computational efficiency for ECG classification. However, limitations include a relatively small dataset and lack of real-time validation. The authors suggest expanding the dataset, testing on multi-center data and developing lightweight versions for real-time clinical use.

The paper by Alghieth (2025) [26] presents a novel hybrid deep learning framework that combines Convolutional Neural Networks (CNNs) and Transformer architectures for accurate and efficient real-time ECG anomaly detection. The model was trained using publicly available ECG datasets — specifically, the MIT-BIH Arrhythmia Database and the PhysioNet 2017 Challenge Dataset, which include diverse arrhythmia types such as Atrial Fibrillation (AF), Premature Ventricular Contractions (PVCs), and Supraventricular Premature Beats (SPBs). DeepECGNet integrates CNN-based local feature extraction with Transformer-based global dependency learning, enhanced by an adaptive anomaly detection module and an

attention-based denoising mechanism for noise resilience. Experimental results show that the model achieves 98.3% accuracy, 97.1% precision, 96.5% recall, and an F1-score of 0.965, outperforming existing models like ECGNet (96.5%) and LSTM-based models (94.2%). The model’s Signal-to-Noise Ratio (SNR) improved from 5.2 dB to 14.5 dB, with Mean Squared Error (MSE) reduced from 0.042 to 0.007, and its low 30 MB memory footprint and 50 ms latency on a Raspberry Pi 4B demonstrate suitability for real-time, wearable deployment. Moreover, the integration of federated learning ensures privacy-preserving distributed training. However, limitations include the model’s reliance on binary classification (normal vs. abnormal), potential generalization challenges across unseen patient populations, and the need for further validation using multi-lead or larger clinical datasets.

2.3 Summary of Key Findings

Deep learning-based CVD prediction has rapidly progressed from manual feature engineering toward robust, multimodal and self-supervised approaches. Early ensemble models like CraftNet, combining R-R intervals and wavelet descriptors, achieved high sensitivity (89.3%) and specificity (95.8%) but depended on labor-intensive feature design. Sparse autoencoders with 2D-CNNs reached 90% accuracy on small ECG sets yet lacked interpretability, while CNN-LSTM frameworks augmented with SHAP explanations hit 74% accuracy but suffered generalizability issues. Incorporating additional modalities—fundus images with clinical risk factors—boosted AUROC to 0.905, though demographic diversity and explainability remained concerns. Fusion of MRI and ECG via 3D U-Net and graph neural networks yielded 93% accuracy but risked overfitting on proprietary data. Lightweight 1D-CNNs reported near-perfect accuracy (99.9%) on tiny heart disease cohorts, limiting real-world applicability. Self-supervised contrastive pretraining on MIMIC-IV ECG achieved AUROC 0.96 using minimal labels, demonstrating data efficiency but constrained to single-lead signals. Larger PTB-XL studies combining SMOTE-balanced hand-crafted features with CNN/DNN attained 84% accuracy, while anomaly detection architectures (FADE) reached 85% accuracy but only distinguished normal vs. abnormal. Future efforts should emphasize transformer-based models trained on diverse, large-scale ECG datasets, integrated with explainable AI techniques for clinically actionable CVD diagnosis.

Table 2.1: Summary of Related Studies on Cardiovascular Disease Prediction Using Multimodal and ECG-Based Models

Study (Year)	Data & Modalities	Model & Metric	Limitation
Li et al. (2020)	MIT-BIH ECG	CraftNet: Sens 89.25%, Spec 95.79%	Manual feature design
Ordás et al. (2023)	Mixed ECG datasets	Autoencoder+2D-CNN: Acc 90.09%	<1,000 samples
Hossain et al. (2023)	Kaggle ECG (70,000)	CNN-LSTM+SHAP: Acc 74.15%	Limited validation
Lee et al. (2023)	Fundus + risk factors	DenseNet+DNN: AUROC 0.905	Poor interpretability
Alsekait et al. (2024)	MRI + ECG	3D U-Net+TCGNN: Acc 92.6–93.5%	Overfitting risk
Honi & Szathmary (2024)	UCI Heart Disease (1,025)	1D-CNN: Acc 99.95%	Tiny dataset
Liu et al. (2024)	ECG + X-ray + text (440 cases)	1D-CNN+DenseNet121: Acc 99.63%	No external testing
Nowroozilarki et al. (2024)	MIMIC-IV ECG	ResNet50 (contrastive): AUROC 0.96	Single-lead only
Nayyab et al. (2025)	PTB-XL ECG (17,232)	CNN/DNN: Acc 81–84%	SMOTE, manual features
Ruiz-Barroso et al. (2025)	MIT-BIH ECG & PCG	FADE: Acc 85.45% (anomaly)	No subtype classification

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

Training and running the deep learning model demanded substantial computing resources. Our approach combined a ResNet architecture for processing ECG signals with an FT-transformer network for handling numeric features. The dataset itself was quite large, containing 550171 rows of 12-channel ECG recordings along with additional numerical attributes. As a result, the hardware requirements were considerable. Sufficient RAM, high-speed storage and a capable GPU were essential to manage the data and train the model efficiently.

3.1.1 Recommended Hardware

For projects of this scale, a reasonably powerful system is necessary:

1. **CPU:** A modern Intel Core i7 or AMD Ryzen 7 processor can reliably manage data preprocessing and model training tasks.
2. **RAM:** A minimum of 64 GB of memory is recommended to handle large batch sizes, data loading operations, and parallel processing without bottlenecks. Working with nearly 560,000 rows, large batch sizes and data loaders can place significant demands on memory—especially when using data augmentation, caching or multiprocessing.
3. **GPU:** A high-memory GPU is essential for efficient training. An NVIDIA RTX 3090 with 24 GB VRAM is considered the minimum. When available, datacenter-grade GPUs such as the A100, H100 or V100 can significantly reduce training time and allow for larger batch sizes and more complex models.
4. **Storage:** A fast SSD with at least 1 TB of free space is recommended. This is necessary to store the preprocessed .npy signal files, feature CSVs, model checkpoints and logs. Using an SSD also ensures faster data loading during training.
5. **Operating system:** A 64-bit operating system is required, with Linux being preferred for its stability, driver support and optimized performance with frameworks such as PyTorch.

3.1.2 Software Requirements

The ECG classification model relies on a Python-based deep learning stack. Python version 3.9 or higher is recommended. The primary libraries include PyTorch (with CUDA support for GPU acceleration), scikit-learn for machine learning utilities, pandas and numpy for data manipulation, and tqdm for progress tracking. These dependencies can be installed using pip (e.g., `pip install torch torchvision scikit-learn pandas numpy tqdm`). In addition, CUDA 11.7+ and cuDNN must be properly configured to enable GPU-based training and inference.

3.1.3 Data Requirements

A sufficiently large labeled dataset is essential for model performance. In this work, a minimum of 0.5 million labeled samples was used for training, validation and testing. During data processing, the ECG signals were stored as .npy files, each containing 12-lead recordings with 5,000 sample points per lead. A corresponding CSV file provides the numeric features and labels. The dataset needs to be preprocessed prior to training, i.e. the null values should be eliminated and the problem of class imbalance should be addressed so that the model learning can be robust and effective.

3.2 Societal Impact

Our study has a significant impact on society, health, safety, legal and cultural aspects. The societal significance of this research lies in its ability to strengthen and improve public health and healthcare services. By detecting normal, borderline and abnormal heart conditions and identifying specific cardiac diseases and heart conditions, the study can help decrease misdiagnosis and delayed treatment which ultimately improves patient outcomes. Healthcare workers benefit from this research because it helps them make informed decisions. It motivates people to stay mindful of their heart health. This research increases awareness about heart disease and preventive care which leads to better and improved health education along with reduced cases of serious problems in heart. This ultimately reduces the weight of cardiovascular diseases in society. Additionally, our approach also offers scalable and technology-driven solutions. These solutions can be used in clinics and hospitals where resources are limited. Thus, the research helps reduce healthcare inequality and increase access to high-quality diagnostics.

Automated ECG signal diagnosis has the ability to transform cardiovascular care and decrease mortality significantly. The impact on cardiac-specific deaths is even more striking, with a significant reduction observed. This improvement is largely due to the technology’s ability to detect subtle ECG changes that may be overlooked by human interpreters. By recognizing these patterns earlier, interventions can be initiated before conditions progress to critical or life-threatening stages. Integrating automated ECG systems with telemedicine platforms creates opportunities for continuous cardiac monitoring. Early-warning alerts for arrhythmias or other emergencies can help prevent thousands of avoidable deaths by enabling timely intervention.

This research also plays a significant role in improving and enhancing patient safety. Early detection of cardiovascular disease helps doctors take quick actions and prevent severe complications such as heart attacks or irregular heart rhythms. The study reduces the chance of diagnostic errors. As a result, this reduces the risk of wrong or delayed treatments which can positively impact patient’s illness and death rates. Furthermore, the use of deep learning models in ECG analysis minimizes the need for manual interpretation. Also, manual interpretations can vary between clinicians. Automation helps standardize care and enhance safety across various healthcare settings. The model contributes to creating a safer environment for patients by ensuring that critical heart conditions are detected reliably and promptly.

By bringing technology into heart disease detection, this research has the power to change how people think about health and illness. The findings highlight the importance of early diagnosis and remind communities to take responsibility for protecting heart health. In many situations, cardiovascular diseases are often overlooked or misunderstood. But this research helps to address that gap. It makes data-driven tools and scientific methods a regular part of health practices. This helps people become more familiar with and comfortable using them. At the same time, it also builds trust in technology-based healthcare. This, in turn, encourages them to accept modern medical innovations together with traditional care. Over time, the research creates a change in culture. People start caring more about their health. They accept scientific methods as part of everyday life.

3.3 Environmental Impact

The environmental impact of this research primarily relates to the computational resources required for training and deploying deep learning models using the MIMIC-IV ECG dataset. Training models such as ResNet for ECG classification involves high-performance hardware, which consumes significant amounts of electricity and contributes to energy use and associated carbon emissions. Servers and data centers are required to store and process large amounts of data. These systems consume electricity and need constant cooling, which increases environmental costs. However, through the development of efficient and accurate models, this research helps reduce repeated tests and unnecessary hospital visits. As a result, this indirectly lowers the environmental load of the healthcare system. Additionally, encouraging the use of automated ECG analysis helps optimize use of resources in clinical environments which reduces the dependency on physical materials like paper-based records and diagnostic kits. While computing systems do have an environmental impact, the overall outcome is positive. The long-term advantage of streamlined diagnostics, resource optimization and decreased medical waste highlight a positive and responsible contribution toward sustainable healthcare practices.

3.4 Ethical Issues

Since our research involves sensitive medical data, protecting and ensuring patient privacy is an important ethical concern. As researchers, treating patient data with

strict confidentiality and following all relevant data use agreements or regulations is our duty. We must use the dataset only for valid research purposes and ensure that no attempt is made to re-identify individuals. Secure storage and careful data processing are also essential. We must also report results honestly, prevent misinterpretation, and test models for fairness and reliability before applying them in practice. The dataset used in the study was approved by the Institutional Review Boards at Beth Israel Deaconess Medical Center in Boston and the Massachusetts Institute of Technology in Cambridge. Since all protected health information was deidentified, the requirement for individual patient consent was waived. Responsible and secure use of patient data is essential. It ensures that individual rights are respected. Also, confidence in medical research remains strong.

3.5 Project Management Plan

This thesis unfolded as an iterative, experiment-driven project, with phases intentionally overlapping so that insights from one stage could immediately inform the next. The plan was flexible by design: methods and compute choices were adjusted along the way to match actual resource constraints, reduce training time and keep model development practical. The schedule shown below in the fig 3.1 reflects the actual workflow which was carefully planned, guided by the data and adapted continuously for improved results.

1. **Week 1-12: Literature review and problem shaping**

The initial three months were dedicated to understanding the field’s current achievements and persistent challenges. Our reading spanned ECG classification, disease prediction, multimodal deep learning models emphasizing evaluation quality and clinical applicability. This review helped define the central problem: developing ECG classifiers at the disease level that could also address class imbalance and the limitations of disease-specific models.

2. **Week 10-16: Data acquisition and preprocessing**

Alongside the reading phase we built the data pipeline. ECGs in WFDB format were processed into 2D NumPy arrays making them ready for GPU training. Alongside this, tabular features such as intervals, amplitudes and axes were saved into CSV files to support multimodal approaches. The preprocessing workflow included normalization, scaling, proper treatment of missing values and a careful train/validation/test split to guarantee fair evaluation.

3. **Week 14-22: Baseline models and first measurements**

Once the data pipeline was in place several baseline models were implemented including DenseNet, ResNet, CNN-BiLSTM and transformer-inspired variants to set reference points for performance. Evaluation focused not only on accuracy but also on precision, recall, F1 scores and other metrics to capture clinically relevance. Initial results suggested that DenseNet and ResNet-style backbones offered the most potential for deeper exploration.

4. **Week 20-24: ResNet build-out and multimodal fusion**

Next, we designed a dedicated ResNet-based classifier specifically for ECG signals, with a parallel branch to integrate tabular features. We systematically explored hyperparameters, including learning rates, batch sizes and optimizers while tracking results by disease category to pinpoint successes and areas needing support. This stage validated the multimodal strategy as an effective way to enhance performance.

5. **Week 24-30: Disease-focused evaluation at scale**

We expanded evaluation to cover specific cardiac conditions including LBBB, RBBB, AV block, tachycardia, bradycardia, arrhythmia, PVC, PAC, left ventricular hypertrophy and MI so we could see how the model behaved in clinically meaningful subsets. Weighted sampling and loss-function adjustments were implemented to mitigate class imbalance, resulting in more stable metrics and improved handling of underrepresented classes.

6. **Week 30-45: Analysis and ablations**

The results phase went deeper than headline accuracy—metrics included Precision, Recall, F1-score, AUROC, AUPRC, Specificity rate to reveal different aspects of utility. Ablation studies compared signal-only models to multimodal variants and isolated the effects of architecture and hyperparameters. These analyses clarified what truly moved the needle.

7. **Week 45-48: Write-up and final polish**

The final stretch focused on communicating the work clearly and completely from. Chapters were drafted and refined, figures and tables finalized and references polished in LaTeX. The documentation closes the loop: it captures the methodology, the experiments and what the results mean for ECG disease classification and risk prediction.

In short, the plan balanced rigor with adaptability: read widely, build strong baselines, iterate with feedback from metrics and finish with careful analysis and clear writing. The overlapping phases ensured that each improvement was quickly tested in context, making steady progress toward accurate and clinically relevant ECG modeling.

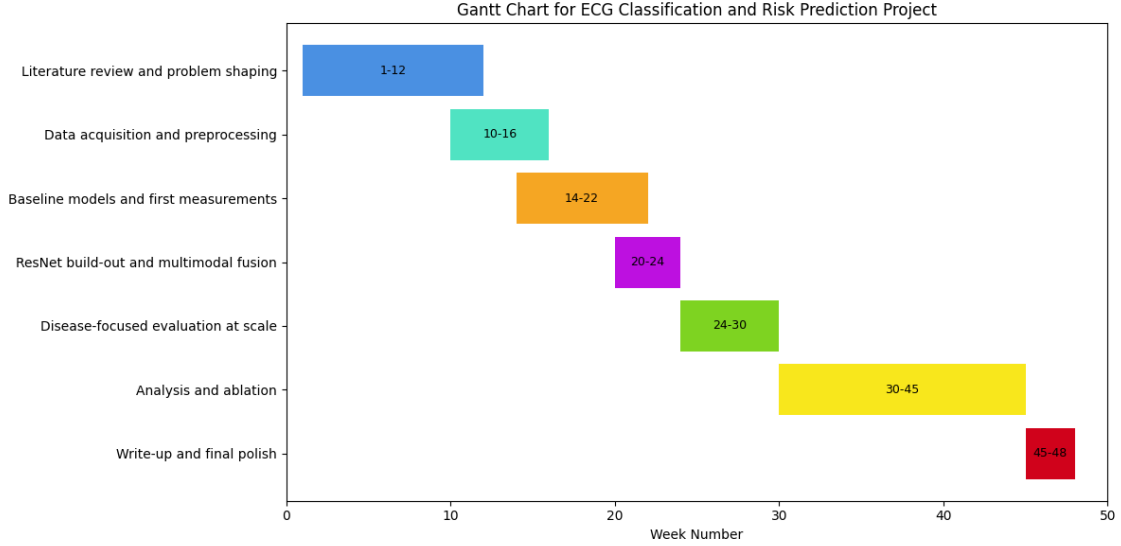


Figure 3.1: Project Management Plan

3.6 Risk Management

Data availability and quality posed a significant risk throughout the project. WFDB-format ECG datasets are not only very large but also frequently contain noise, incomplete records and corrupted values that can undermine model performance. To address this, we first converted all signals into NumPy arrays for efficient handling. We then rigorously filtered out any samples with missing or obviously corrupted data and applied feature scaling to ensure uniformity across every record. Another critical risk was class imbalance among disease categories. Rare conditions like AV block appeared far less often than more common diagnoses such as myocardial infarction, creating the danger that our models would favor majority classes during training. To counteract this, we implemented weighted random sampling to give underrepresented classes more presence in each batch, applied focal loss to focus learning on harder-to-classify examples and used data augmentation to artificially expand the minority class samples. Together, these strategies promoted fairer and more balanced model training. Overfitting also emerged as a technical risk especially given the high representational capacity of architectures like DenseNet and ResNet. These models can tend to memorize limited training data rather than generalize to new cases. We mitigated this by introducing dropout layers to reduce co-adaptation of neurons, incorporating batch normalization to stabilize learning and employing cross-validation to rigorously test model generalization. Computational resource limitations represented another major challenge, particularly when training transformer-based and multimodal models that demanded extensive GPU memory and long training times. To manage these constraints we optimized batch sizes to fit our hardware, adopted mixed-precision training to reduce memory usage and speed up computation and leveraged high-performance computing clusters to distribute the workload efficiently. From a project management standpoint, the risk of scope creep was ever-present: potential expansions into multimodal feature integration, disease-level classification and comprehensive risk scoring could have derailed the original objectives. We controlled this by defining and documenting the project scope early focusing our efforts on ECG classification integrated with

risk analysis and regularly reviewing milestones to ensure alignment. Finally, ensuring reproducibility and validation was crucial for maintaining research credibility. We established strict version control over datasets, pre-processing scripts, model architectures, hyperparameters and evaluation protocols using GitHub. Detailed experiment logs captured every configuration and result enabling other researchers to reproduce our findings exactly. By proactively identifying these risks and deploying structured mitigation strategies we were able to keep the project on track, deliver technically robust models and uphold the scientific integrity of our work.

3.7 Economic Analysis

This thesis’s economic analysis compares the cost-effectiveness of a deep learning based ECG prediction and risk stratification system to traditional diagnostic methods. The primary expenses were computational resources, data processing and research infrastructure. Handling large ECG datasets required substantial storage, preprocessing efforts and GPU acceleration for training ResNet and transformer-based models. Whether using cloud services or high-performance local hardware resource costs were significant. Indirectly, the time and expertise of research personnel for tasks like data cleaning, model experiments and evaluations also represented a major investment. The project budget shown in table 3.1 was designed to balance performance and cost-effectiveness.

Table 3.1: Budget Planning of the Project

Category	Allocation (BDT)	Description
Personnel	200,000	Graduate assistant support (6 months)
Hardware	100,000	GPU rental or modest workstation upgrade
Cloud Services	50,000	GPU instance hours and storage for 6 months
Research & Publication	50,000	Conference fees and open-access charges
Data & Preprocessing	50,000	CFET toolkit development and storage backup
Operations & Misc	100,000	Connectivity, lab access, and contingency
Total	550,000	

On the benefit side, the system offers clear clinical and economic advantages. Automated ECG classification speeds up diagnosis and reduces cardiologists’ workload by quickly detecting and categorizing conditions such as LBBB, RBBB, AV block, tachycardia, bradycardia, arrhythmia, PVC, PAC and myocardial infarction. This not only accelerates patient care but also cuts the cost of manual interpretation. Adding risk scoring further supports decision-making by helping providers prioritize treatment based on severity. The system can lower misdiagnosis rates, avoid unnecessary tests and improve early detection, all of which reduce long-term treatment expenses.

When weighing the initial investment against these projected benefits the cost-benefit ratio clearly favors real-world adoption. After deployment, the system needs minimal maintenance and can scale to process large volumes of ECG data efficiently. Over time, the improved patient outcomes, fewer diagnostic errors and time savings for healthcare providers far outweigh the upfront computational and research costs.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

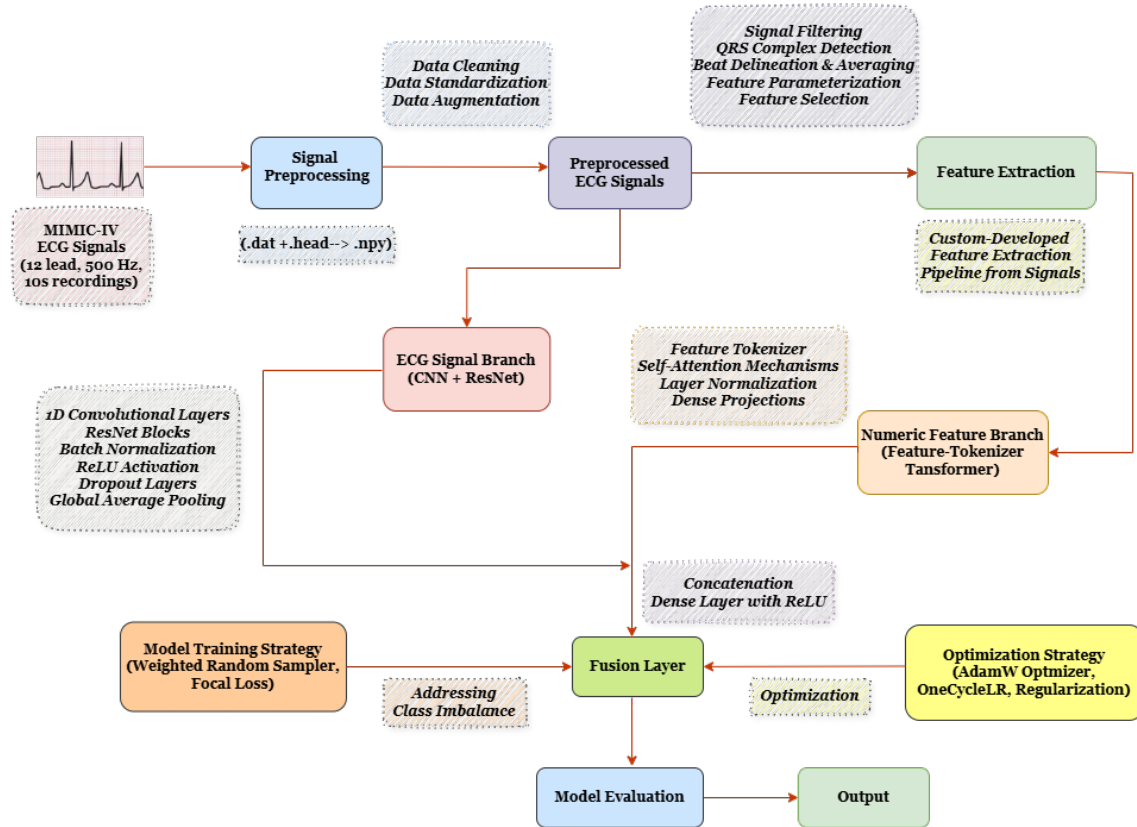


Figure 4.1: Overview of the Proposed Multimodal Deep Learning Framework Combining ECG and Numerical Features for Cardiovascular Disease Classification.

The main aim of this study is to establish a multimodal deep learning model to combine the raw electrocardiogram (ECG) recordings along with the structured numeric clinical characteristics to properly separate cardiovascular risks. The approach is a quantitative, data-driven methodology that is appropriate to the inherent nature of the data: time-series ECGs and tabular numeric data and also meets technical and clinical constraints such as imbalance between classes, large volume of data and finite resources on a GPU. The design provides accurate classification, effective training

and future generalization to clinical practice in the real world. The methodology adopted in this study follows a mixed approach, integrating both quantitative and qualitative elements to achieve a comprehensive understanding of cardiovascular disease prediction using ECG data. The quantitative component forms the foundation of the research, as the model was trained and evaluated on large-scale numerical datasets containing ECG signal arrays and extracted clinical features such as QT interval, QRS duration and heart rate. Advanced quantitative techniques, including weighted random sampling, focal loss and adaptive optimization (AdamW with OneCycleLR scheduling) were employed to address class imbalance and enhance model generalization. Model performance was assessed using statistical and numerical metrics such as accuracy, precision, recall, specificity, F1-score, AUROC and AUPRC ensuring objective measurement and reproducibility. Complementing this, the qualitative aspect of the methodology lies in the theoretical reasoning behind architectural design choices, feature selection and interpretation of model outcomes in the context of clinical significance. This interpretive layer helped explain why certain architectures (e.g., ResNet and Transformer) or regularization techniques were selected and how model predictions relate to real-world diagnostic value. Hence, the methodology effectively combines data-driven quantitative rigor with qualitative clinical and theoretical reasoning, ensuring both computational robustness and practical relevance in cardiovascular disease classification.

4.1.1 Objective Formulation

The first phase involves defining the problem and setting clear objectives. The primary goal was to design a dual-input deep learning model capable of handling two data modalities. Firstly, time-series ECG signals which contain rich temporal patterns indicative of cardiac activity. Secondly, numeric clinical features which are extracted from ECG signals using CardioFeature Extraction Toolkit (CFET), including QT interval, QRS duration, PR interval, P-wave duration, corrected QT (QTc), RR interval and Heart Rate. The dual-input approach leverages complementary information: ECG signals provide temporal dynamics of heart function, while numeric features supply clinical context. The model needed to balance high predictive performance with practical constraints, including class imbalance, large data volume and GPU memory limitations.

4.1.2 Data Acquisition and Preprocessing

The data were collected from the MIMIC-IV ECG dataset [11], which included raw .dat and .hea files along with patient metadata. To optimize training efficiency ECG signals were converted to .npy format shown in Fig. 4.1, enabling fast on-the-fly loading during training. Numeric features were extracted using CardioFeature Extraction Toolkit (CFET) and stored together with the corresponding .npy signal paths, labels and disease annotations in a single CSV file. Each row of this CSV contained the path to the ECG signal, numeric features, the overall ECG label (Normal, Abnormal, or Borderline) and disease-specific labels including PAC, PVC, MI, Tachycardia, Bradycardia, A-V block, LBBB, RBBB, Arrhythmia and Left Ventricular Hypertrophy. Preprocessing steps included standardizing numeric features using StandardScaler and handling missing values in both numeric features

and ECG signals using zero imputation. For training, data augmentation such as additive Gaussian noise, signal scaling and temporal shifts was applied to improve generalization. The dataset was then stratified and split into training, validation and test sets to maintain class distributions.

4.1.3 Model Architecture Design

The proposed model CardioFusion-RT adopts a dual-input deep learning architecture designed to effectively integrate multimodal information from two complementary sources: the raw 12-lead ECG signal recordings and the tabular numeric features derived from feature extraction tools. The motivation behind this design is to leverage the high temporal resolution of ECG waveforms alongside clinically relevant hand-crafted features to achieve a more robust and accurate diagnostic model. The model is structured into three primary components: the ECG signal branch, the numeric feature branch and the fusion layer that combines the learned representations into a final decision module. The whole model architecture is explained below following the Fig. 4.2.

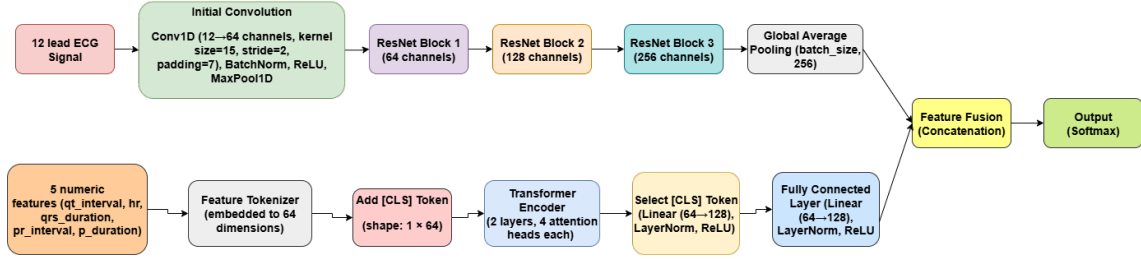


Figure 4.2: Model Diagram

4.1.3.1 ECG Signal Branch

This branch is specifically aimed to process raw electrocardiogram (ECG) waveforms which are sequential time series data defined by complex time and spatial relationships. This line starts by using one-dimensional convolutional layers, which are especially effective in time-series analysis as the local temporal relationships are modeled with the help of learnable filters operating on short portions of the signal. These convolutional layers allow the network to extract low-level features which include peaks, slopes and local features which are associated with notable physiological events in the ECG including the QRS complex, P wave and T wave. Residual Network (ResNet) blocks are used to improve feature learning to a deeper level. ResNet also presents residual connections, which avoid intermediate layers and thus, the model is able to learn to map identities when no additional transformations are required. The architecture design helps to address the widespread issue of gradient disappearance in deep networks and it allows training much deeper architecture. In the view of the ECG analysis, when the low-level morphology and higher-level interactions of waves are both consequential, residual connections guarantee the preservation of both simplistic and complex representations. Each ResNet block also integrates Batch Normalization (BN) and ReLU activation functions. Batch Normalization stabilizes the learning process by normalizing intermediate feature maps, reducing internal covariate shift, and enabling the use of higher learn-

ing rates. ReLU (Rectified Linear Unit) is chosen for its computational efficiency and ability to introduce non-linearity, which allows the network to learn highly non-linear signal transformations essential for capturing arrhythmias or subtle waveform abnormalities. To enhance generalization and prevent overfitting, Dropout layers are introduced after convolutional operations. Dropout randomly deactivates a fraction of neurons during training, encouraging redundancy and robustness in feature representation. Given the high variance often present in physiological signals, dropout ensures that the model does not memorize specific patterns from the training set but instead learns more generalizable representations. Finally, Global Average Pooling (GAP) is applied before the fusion stage. GAP reduces the dimensionality of feature maps by computing the average across time, effectively summarizing temporal features into a compact vector representation. This not only reduces the number of trainable parameters (compared to fully connected layers) but also makes the model invariant to minor temporal shifts in the ECG signals, which are common in real-world clinical settings.

4.1.3.2 Numeric Feature Branch

Parallel to the signal branch, the second branch is designed to handle tabular numeric features that were extracted from the ECG recordings using CardioFeature Extraction Toolkit (CFET) using Multi-Lead Beat Averaging and FeatureDB toolkit. These features include clinically relevant attributes such as QT interval, QRS duration, PR interval, P duration and heart rate variability measures. While deep learning excels at automatically learning representations from raw signals, these carefully engineered features encapsulate domain knowledge and provide complementary insights that may not be easily captured from the raw data alone. To process this modality, we employ a Feature Tokenizer Transformer (FT-Transformer), which is an adaptation of transformer architectures specifically optimized for tabular data. In this branch, each numeric feature is treated as a “token” and the transformer applies self-attention mechanisms to model interdependencies between different features. For instance, the clinical interpretation of QT interval is often dependent on the heart rate (Bazett’s formula for QTc correction), and self-attention enables the model to automatically learn such relationships without manual feature engineering. The FT-Transformer uses Layer Normalization instead of Batch Normalization, which is more effective for non-sequential tabular data and ensures stable learning dynamics. Activation functions such as ReLU are incorporated to introduce non-linearity, allowing the branch to capture complex feature interactions beyond linear correlations. Moreover, fully connected layers following the transformer blocks compress the learned embeddings into a dense numeric representation that aligns dimensionally with the signal branch for effective fusion.

4.1.3.3 Fusion Layer

After both branches independently process their respective inputs, their outputs are concatenated to form a joint representation that combines temporal-spatial features from ECG waveforms with relational embeddings of numeric clinical features. This fusion strategy is deliberately chosen because concatenation preserves the full richness of both modalities before they are projected into a shared latent space. The concatenated vector is then passed through additional fully connected layers with

ReLU activation and dropout regularization. These layers serve as a decision-making module, where the network learns to exploit complementary relationships between modalities. For example, a prolonged QRS duration detected in the signal branch may reinforce abnormalities identified in the numeric branch, such as left bundle branch block (LBBB). Dropout at this stage ensures robustness against overfitting, especially important given the multimodal complexity of the data. Finally, the model concludes with a softmax classification layer, which outputs the probability distribution across diagnostic categories. The selected categories include both general ECG states—Normal, Borderline and Abnormal as well as specific disease-related conditions representing a wide range of cardiac abnormalities. Softmax ensures that probabilities sum to one, allowing for interpretable outputs that can guide clinical decision-making. The algorithm of the whole model is given below: P

Algorithm 1 CardioFusion-RT for Disease Classification

Input: ECG_signal $\in \mathbb{R}^{B \times 12 \times 5000}$, Numeric_features $\in \mathbb{R}^{B \times 5}$

Output: Binary probability for disease classification (positive class)

- 1 **Main Architecture:** (1) ECG signal branch (ResNet-1D) ; (2) Numeric features branch (FT-Transformer) ; (3) Feature fusion & classifier.
-

Algorithm 2 RESNET_ECG_BRANCH(ECG_signal)

Input: ECG_signal $\in \mathbb{R}^{B \times 12 \times 5000}$

Output: signal_features $\in \mathbb{R}^{B \times 256}$

- 2 conv1 $\leftarrow \text{Conv1D}(12 \rightarrow 64, k=15, s=2, p=7)$ bn1 $\leftarrow \text{BatchNorm1D}(64)$,
 relu $\leftarrow \text{ReLU}()$, pool $\leftarrow \text{MaxPool1D}(k=3, s=2, p=1)$ x $\leftarrow$
 pool(relu(bn1(conv1(ECG_signal))))
 3 x $\leftarrow \text{RESNET_LAYER1}(x)$ // 64 $\rightarrow$ 64, 2 blocks
 4 x $\leftarrow \text{RESNET_LAYER2}(x)$ // 64 $\rightarrow$ 128, 2 blocks
 5 x $\leftarrow \text{RESNET_LAYER3}(x)$ // 128 $\rightarrow$ 256, 2 blocks
 6 x $\leftarrow \text{AdaptiveAvgPool1D}(1)(x)$ signal_features $\leftarrow \text{Flatten}(x)$ // [B, 256]
 7 **return** signal_features
-

Algorithm 3 MAKE_RESNET_LAYER($C_{\text{in}}, C_{\text{out}}, \text{num_blocks}, K, S$)

Output: Sequential of RESBLOCK1Ds

- 8 **if** $S \neq 1$ **or** $C_{\text{in}} \neq C_{\text{out}}$ **then**
 9 down $\leftarrow \text{Sequential}(\text{Conv1D}(C_{\text{in}} \rightarrow C_{\text{out}}, k=1, s=S), \text{BatchNorm1D}(C_{\text{out}}))$
 10 **else**
 11 down $\leftarrow \text{None}$
 12 layers $\leftarrow [\text{RESBLOCK1D}(C_{\text{in}}, C_{\text{out}}, K, S, \text{down})]$ **for** $i \leftarrow 2$ **to** num_blocks **do**
 13 append RESBLOCK1D($C_{\text{out}}, C_{\text{out}}, K, 1, \text{None}$) to layers
 14 **return** Sequential(layers)
-

Algorithm 4 RESBLOCK1D($x, C_{\text{in}}, C_{\text{out}}, K, S, \text{down}$)

Input: $x \in \mathbb{R}^{B \times C_{\text{in}} \times L}$ **Output:** $\in \mathbb{R}^{B \times C_{\text{out}} \times L/S}$

```
15 res  $\leftarrow x$    $y \leftarrow \text{Conv1D}(C_{\text{in}} \rightarrow C_{\text{out}}, k=K, s=S, p=(K-1)/2)(\text{ReLU}$   
     $(\text{BatchNorm1D}(C_{\text{in}})(x)))$    $y \leftarrow \text{Conv1D}(C_{\text{out}} \rightarrow$   
     $C_{\text{out}}, k=K, s=1, p=(K-1)/2)(\text{ReLU}(\text{BatchNorm1D}(C_{\text{out}})(y)))$   if down  $\neq \text{None}$   
    then  
16    $\text{res} \leftarrow \text{down}(x)$   
17 return  $y + \text{res}$ 
```

Algorithm 5 FT_TRANSFORMER_BRANCH(numeric_features)

Input: numeric_features $\in \mathbb{R}^{B \times 5}$ **Output:** num_emb $\in \mathbb{R}^{B \times 128}$

```
18 Set:   embed_dim=64,   heads=4,   layers=2,   out_dim=128    $\tilde{X} \leftarrow$   
     $\text{Unsqueeze}(\text{numeric\_features}, -1)$     $// [B, 5, 1]$   
19  $T \leftarrow \text{Linear}(1 \rightarrow 64)(\tilde{X})$     $// [B, 5, 64]$   
20  $\text{CLS} \in \mathbb{R}^{1 \times 1 \times 64}$  learnable;  $\text{CLS}_B \leftarrow \text{Expand}(\text{CLS}, B, -1, -1)$   $Z \leftarrow$   
     $\text{Concat}([\text{CLS}_B, T], \text{dim}=1)$     $// [B, 6, 64]$   
21 for  $\ell \leftarrow 1$  to layers do  
22    $Z \leftarrow \text{TRANSFORMERENCODERLAYER}(Z, 64, 4, 128)$   
23  $h \leftarrow Z[:, 0, :]$     $// \text{CLS}$   
24  $h \leftarrow \text{Linear}(64 \rightarrow 128)(\text{ReLU}(\text{LayerNorm}(64)(h)))$  return num_emb  $\leftarrow h$ 
```

Algorithm 6 TRANSFORMERENCODERLAYER(X, d_m, n_h, d_f)

Input: $X \in \mathbb{R}^{B \times L \times d_m}$

```
25  $A \leftarrow \text{MultiHeadAttention}(d_m, n_h)(X, X, X)$   $X \leftarrow \text{LayerNorm}(d_m)(X + A)$   
     $F \leftarrow \text{Linear}(d_m \rightarrow d_f)(X)$ ;  $F \leftarrow \text{Linear}(d_f \rightarrow d_m)(\text{ReLU}(F))$   $X \leftarrow$   
     $\text{LayerNorm}(d_m)(X + F)$  return  $X$ 
```

Algorithm 7 FEATURE_FUSION_CLASSIFIER(s, n)

Input: $s \in \mathbb{R}^{B \times 256}, n \in \mathbb{R}^{B \times 128}$

```
26  $u \leftarrow \text{Concat}([s, n], \text{dim}=1)$     $// [B, 384]$   
27  $u \leftarrow \text{Dropout}(0.5)(\text{ReLU}(\text{Linear}(384 \rightarrow 256)(u)))$  return  $\text{Linear}(256 \rightarrow 2)(u)$   
     $// \text{logits}$ 
```

Algorithm 8 FORWARD_PASS(ECG_signal, numeric_features)

 $s \leftarrow \text{RESNET_ECG_BRANCH}(\text{ECG_signal})$ $n \leftarrow \text{FT_TRANSFORMER_BRANCH}(\text{numeric_features})$ $\text{FEATURE_FUSION_CLASSIFIER}(s, n)$

Algorithm 9 TRAINING_PROCEDURE

Init: parameters θ ; AdamW (lr=10⁻³, wd=10⁻²); Focal Loss ($\gamma=1.5$); early stopping (patience=10, min_Δ=10⁻³); best_val_F1 ← 0 **for** $epoch = 1$ **to** MAX_EPOCHS **do**

```
    // Training
28   set model to train foreach (ECGb, numb, yb) in train_loader do
29     if yb = 1 and rand() < AUGMENT_PROB then
30       ECGb ← APPLY_AUGMENTATION(ECGb)
31     zero optimizer grads  $z \leftarrow$  FORWARD_PASS(ECGb, numb)  $L \leftarrow$ 
       FocalLoss( $z, y_b$ ) backpropagate  $L$ ; optimizer step; scheduler step
    // Validation
32   set model to eval; collect  $\hat{y}, y$  foreach (ECGv, numv, yv) in val_loader do
33     no-grad:  $z_v \leftarrow$  FORWARD_PASS(ECGv, numv); append arg max( $z_v$ ) and
        $y_v$ 
34   F1 ← F1_SCORE( $y, \hat{y}$ ) if F1 > best_val_F1 then
35     save checkpoint; best_val_F1 ← F1
36   update early stopping with F1; if stopping then
37     break
return trained model
```

Algorithm 10 APPLY_AUGMENTATION(X)

Input: $X \in \mathbb{R}^{B \times 12 \times 5000}$

```
38  $X \leftarrow X + \mathcal{N}(0, 0.01^2)$  // additive Gaussian noise
39  $X \leftarrow X \times \mathcal{U}(0.9, 1.1)$  // amplitude scaling
40  $X \leftarrow Roll(X, shift = \mathcal{U}\{-10, \dots, 10\}, axis=time)$  return  $X$ 
```

Algorithm 11 INFERENCE_WITH_THRESHOLD(model, ECG, num, τ)

```
set model to eval no-grad:  $z \leftarrow$  FORWARD_PASS(ECG, num)  $p \leftarrow$ 
Softmax( $z, dim = 1$ )[:, 1] // positive probability
41  $\hat{y} \leftarrow \mathbb{I}[p > \tau]$  return ( $\hat{y}, p$ )
```

Table 4.1: Model Specifications and Hyperparameters

Signal length	5000 samples
ECG channels	12
Batch size	128
Max epochs	50
Optimizer	AdamW (lr = 10 ⁻³ , weight decay = 10 ⁻²)
Scheduler	OneCycleLR
Loss	Focal Loss ($\gamma = 1.5$)
Dropout	0.5
Augmentation prob.	0.5
Transformer	$d_{\text{model}} = 64$, heads= 4, layers= 2
Output	2 logits → probability via softmax

- **ECG Branch:** Conv1D $\rightarrow$ 3 ResNet layers $\rightarrow$ AdaptiveAvgPool1D $\rightarrow$ flatten $[B, 256]$.
- **Numeric Branch:** Tokenize $[B, 5, 1] \rightarrow [B, 5, 64]$; prepend CLS; Transformer encoder $\times 2$; FC to $[B, 128]$.
- **Fusion:** Concat $[256] \oplus [128] \rightarrow [384] \rightarrow \text{FC}(256) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{FC}(2)$.
- **Output:** Binary probability for disease classification.

4.1.4 Model Training Strategy

4.1.4.1 Training Strategy and Optimization

The training process of CardioFusion-RT was carefully designed to handle the challenges posed by imbalanced classes, high-dimensional ECG signals, multimodal fusion and computational efficiency requirements. Our model was trained for 50 epoches with a batch size of 128. A combination of specialized loss functions, sampling strategies, optimizers, schedulers and regularization techniques were employed to ensure both stable convergence and improved generalization.

4.1.4.2 Addressing Class Imbalance

Clinical ECG datasets are often highly imbalanced, where certain pathological conditions such as Left Bundle Branch Block (LBBB) or Premature Atrial Contractions (PAC) appear much less frequently compared to normal sinus rhythms or other common arrhythmias. Such imbalance poses a challenge, as a naïve model tends to be biased towards the majority class, achieving high overall accuracy while failing to detect rare but clinically significant cases. To address this challenge, two complementary strategies were employed: weighted random sampling and focal loss.

Weighted Random Sampler

To effectively handle the significant class imbalance present in clinical ECG datasets, a Weighted Random Sampling strategy [1] was implemented during the training phase. Instead of selecting samples uniformly at random, each data instance was assigned a probability of being selected that was inversely proportional to the frequency of its class. This means that classes with fewer samples, such as AV-block or PAC, were given higher sampling probabilities, while dominant classes like Normal rhythms were assigned lower probabilities. The process began by calculating the number of samples in each class, followed by determining a corresponding weight for each class as the inverse of its frequency. These class weights were then propagated to individual samples so that every instance carried a specific sampling weight depending on the class it belonged to. During mini-batch formation, the weighted random sampler utilized these probabilities to ensure that underrepresented classes were more frequently drawn, even though the overall dataset distribution remained unchanged. By allowing replacement during sampling, the model could repeatedly encounter minority-class samples across multiple epochs, maintaining balanced exposure throughout training. During data loading, the WeightedRandomSampler

uses these computed weights to choose which samples form each batch. Instead of random uniform selection, it draws samples according to their weight probabilities, meaning:

- Minority class samples (high weights) are selected more frequently.
- Majority class samples (low weights) are selected less frequently.

Over many epochs, this dynamic sampling ensures that the model shows roughly the same number of samples from each class — even though the original dataset distribution is highly skewed. Mathematically, the probability of selecting sample i belonging to class c is:

$$P(i) = \frac{w_c}{\sum_{j=1}^N w_{ij}} \quad (4.1)$$

where $w_c = \frac{1}{N_c}$ and y_j is the class label of sample j .

Hence, if N_c (class count) is small, w_c is large, and $P(i)$ is higher, making the class more represented in training batches.

This weighted sampling technique acts as a data-level balancing strategy that complements loss-level methods (like Focal Loss or class-weighted CrossEntropy). It ensures that:

- Each batch includes diverse class samples
- The model doesn't overfit to the majority rhythm type
- Training remains statistically fair across all arrhythmia categories

Ultimately, it allows the neural network to learn robust and discriminative representations for both common and rare ECG abnormalities, leading to more clinically reliable predictions.

Focal Loss

Beyond sampling and weighting strategies, the choice of loss function was critical in addressing imbalance. Focal loss, an extension of the traditional cross-entropy loss, was employed. It modifies the loss contribution of each sample based on how easily the model classifies it. The focal loss is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (4.2)$$

where p_t is the predicted probability of the true class, α_t is an optional weighting factor for balancing classes, and γ (the focusing parameter) controls how strongly the loss focuses on hard-to-classify examples. When $\gamma = 0$, focal loss reduces to standard cross-entropy. With higher γ such as $\gamma = 1.5$ which was used in our model, correctly classified (easy) samples are down-weighted, while misclassified (hard) samples receive larger penalties. This property makes focal loss particularly effective in highly imbalanced medical datasets, as it compels the model to pay greater attention to underrepresented and misclassified arrhythmias.

WeightedRandomSampler (to rebalance training batches) and focal loss (to emphasize hard examples), the proposed training strategy provided a comprehensive solution to the imbalance problem. Together, these mechanisms ensured that the model achieved not just high accuracy but also improved recall and precision for rare yet clinically important ECG abnormalities.

4.1.4.3 Optimization Algorithm – AdamW

The model was optimized using the AdamW optimizer, a modern variant of Adam that decouples weight decay from the gradient-based parameter updates. Traditional Adam applies L2 regularization by adding the weight decay term directly to the gradient, which can lead to suboptimal regularization. AdamW corrects this by applying weight decay as a separate step, ensuring more effective control over overfitting while retaining the adaptive learning rate benefits of Adam.

AdamW computes individual learning rates for each parameter by maintaining running averages of both the first-order moment (mean of gradients) and the second-order moment (uncentered variance of gradients). This adaptive mechanism allows the optimizer to perform well on sparse or noisy gradients, which are common in ECG time-series data. In practical terms, AdamW accelerates convergence, handles non-stationary patterns in biomedical signals and provides robust performance without extensive hyperparameter tuning.

4.1.4.4 Learning Rate Scheduling – OneCycleLR

To enhance training stability, the OneCycleLR scheduler [5] was used in this study. This scheduler works by gradually increasing the learning rate during the early phase of training, reaching a peak and then slowly decreasing it for the rest of the cycle. The initial warm-up phase helps prevent unstable updates when the model is just beginning to learn, while the higher learning rates that follow encourage the optimizer to explore the loss landscape and avoid getting stuck in sharp local minima. In the final phase, the learning rate is reduced to allow the model to converge smoothly and precisely near the global minimum. This cyclical adjustment of the learning rate not only speeds up convergence but also improves the model’s ability to generalize, making it particularly effective for training complex multimodal architectures such as the one developed in this research.

4.1.4.5 Regularization and Generalization Control

To minimize overfitting, several strategies were incorporated into the training process:

Dropout Regularization

A key method to prevent overfitting was the use of dropout layers throughout the model. Dropout was applied after major convolutional blocks and before the fully connected layers to stop the network from memorizing training data. By randomly turning off some neurons during training, the model learned more general and reliable features. In our implementation, three dropout layers were used:

- **After the first dense block:** Dropout probability set to 0.3, encouraging moderate regularization early in feature extraction.
- **After the second dense block:** Dropout probability set to 0.4, slightly higher due to increased feature complexity at deeper levels.
- **Before the final fully connected classification layer:** Dropout probability set to 0.5, providing stronger regularization at the decision stage where overfitting risk is highest.

This progressive scaling ensured that feature representations remained robust while avoiding excessive information loss in early stages. Dropout introduces stochasticity by randomly “turning off” a fraction of neurons during training, forcing the network to learn redundant pathways for the same representation. Mathematically, during training, each neuron has a probability p of being dropped, while during inference the full network is used with activations scaled by $(1-p)$ to maintain output consistency.

The chosen dropout configuration balanced underfitting and overfitting: lower probabilities in earlier layers allowed stable feature extraction, while higher values in later layers acted as stronger regularizers for the final decision-making process.

4.1.4.6 Early Stopping

To prevent overfitting and ensure generalization, an early stopping strategy was employed. Instead of monitoring validation accuracy—which may be misleading in the presence of class imbalance—the validation F1-score was chosen as the monitoring metric, as it provides a more balanced reflection of precision and recall across all classes. The following parameter settings were used:

- **Monitor Metric:** Validation F1-score (macro-averaged across classes)
- **Patience: 10 epochs** — training was terminated if no improvement in F1-score was observed for 10 consecutive epochs
- **Minimum Delta (min_delta): 0.001** — an improvement in F1 was considered valid only if the change exceeded this threshold, preventing premature stopping due to minor fluctuations
- **Mode: ‘max’** — since higher F1 indicates better performance
- **Restore Best Weights:** Enabled — at the end of training, the model automatically reverted to the checkpoint with the highest validation F1-score rather than the final epoch, ensuring that the best-performing model was retained.

This configuration struck a balance between giving the model enough time to converge and avoiding unnecessary overfitting. Monitoring F1-score instead of accuracy was especially important in this study, as accuracy could remain high even if the model failed to correctly identify minority-class events (e.g., PAC, LBBB).

4.1.4.7 Best-performing Model Checkpoint

The best model was saved based on the highest validation F1-score, ensuring that evaluation used the most balanced and generalizable version rather than one that might have been overfitted later in training. Checkpoints automatically updated whenever a better score was found, working alongside early stopping to keep the most reliable model.

Overall, the training process was designed for both efficiency and clinical accuracy. Class imbalance was handled using focal loss and weighted sampling, while AdamW with OneCycleLR improved training stability and speed. Mixed-precision training optimized GPU use and early stopping prevented overfitting. Together, these steps produced a robust, efficient and clinically reliable ECG classification model.

4.1.5 Testing methodology

The testing methodology of this study was carefully designed to ensure that the evaluation of the multimodal deep learning model was both statistically rigorous and clinically meaningful. The dataset was divided systematically and hyperparameters along with classification thresholds were tuned to avoid bias and enhance generalizability. Instead of using a fixed probability cutoff (such as 0.5), a **dynamic threshold optimization** approach was applied where thresholds ranging from 0.1 to 0.9 were evaluated on the validation set. For each threshold, precision, recall, and F1-score were computed, and the threshold maximizing the F1-score was selected as optimal. This strategy ensured a balanced trade-off between precision—reducing false positives and recall—minimizing false negatives—which is essential in clinical applications. Optimizing for F1 rather than accuracy emphasized both diagnostic reliability and safety, allowing the model to maintain sensitivity for detecting true cardiovascular abnormalities without generating excessive false alarms, thereby aligning the evaluation process with real-world diagnostic priorities.

In summary, the methodology adopted in this study provides a robust, interpretable and practically feasible framework for multimodal ECG classification. By combining high-dimensional ECG signals with structured numeric features and addressing challenges such as class imbalance, missing data and computational constraints, the proposed workflow advances automated, reliable and personalized cardiac risk classification.

4.2 Preliminary Design or Design (Model) Specification

In the initial phase of model development, a DenseNet-inspired 1D convolutional network was implemented to handle raw ECG signals. DenseNet architectures are known for their efficient feature reuse and strong gradient flow, which theoretically makes them suitable for extracting intricate temporal patterns from ECG signals. However, during preliminary simulations, this model exhibited slow convergence and high GPU memory usage, limiting its scalability for large datasets. While it achieved reasonable accuracy on majority classes, its performance on minority classes such as

A-V block and LBBB was suboptimal due to insufficient focus on clinically important patterns, highlighting the need for a more balanced and multimodal approach.

The second design considered was a hybrid CNN-Transformer model, where the CNN branch extracted local temporal features from ECG signals and a transformer-based encoder captured long-range dependencies. This architecture improved the network’s ability to detect subtle temporal variations and complex interactions within ECG signals. Preliminary testing demonstrated faster learning and better sensitivity to minority classes compared to DenseNet, yet the model faced challenges in integrating numeric clinical features effectively, and transformer layers added additional computational overhead, which limited batch size and slowed overall training on the full dataset.

The third alternative was a CNN-BiLSTM model, combining convolutional layers for feature extraction with bidirectional LSTM layers to capture sequential dependencies in ECG signals. This design successfully modeled temporal dynamics and achieved good F1-scores on majority classes. However, BiLSTM layers significantly increased training time and GPU memory consumption, making the approach less practical for large-scale datasets. Moreover, the model struggled to fully leverage tabular numeric features alongside signal data, limiting the benefit of multimodal learning.

The fourth design explored a ResNet-based CNN combined with an MLP for numeric features, where residual connections facilitated deeper networks and mitigated vanishing gradient problems, while the MLP processed tabular clinical data. This architecture was more efficient than the previous models and allowed for direct fusion of multimodal features. Preliminary simulations showed improved generalization and better sensitivity to minority classes, yet the MLP lacked the capability to capture complex interactions among numeric features compared to transformer-based methods, which slightly restricted overall classification performance.

Finally, the CardioFusion-RT model was adopted as the fixed, final architecture. In this design, the ResNet1D branch effectively extracted temporal and spatial features from 12-lead ECG signals, while the FT-Transformer branch encoded numeric clinical features, learning inter-feature relationships and emphasizing clinically relevant attributes. Preliminary simulations on a subset of the dataset demonstrated faster convergence, higher F1-scores across both majority and minority classes, and superior generalization on validation data compared to all previous alternatives. Additional enhancements, including weighted random sampling, focal loss, and data augmentation, further optimized performance and addressed class imbalance, making this design the most robust and scalable solution for large-scale multimodal ECG classification.

Simulations were conducted through training and evaluating the models on the ECG dataset in order to verify the performance and effectiveness of each proposed model. Each model architecture was trained properly using stratified data splits and evaluation are performed based on accuracy, precision, recall and F1 score. These metrics provide a complete picture of categorization performance. The simulation

results validated each model’s respective strengths and limitations which influenced the ultimate model decision.

4.3 Data Collection

For this research purpose, we used Physionet MIMIC-IV ECG dataset [11]. Initially, the MIMIC-IV-ECG module comprises around 800,000 diagnostic electrocardiograms (ECGs), recorded from approximately 160,000 distinct patients. Each ECG is a 10-second recording captured using the standard 12-lead configuration and sampled at a frequency of 500 Hz. Diagnostic ECGs are recorded at Beth Israel Deaconess Medical Center (BIDMC) as part of regular care. In our dataset, 3 types of information are present. (i) The electrocardiogram waveforms themselves, (ii) the machine measurements (ex: average RR interval as calculated by the machine) and (iii) the cardiologist reports.

For each diagnostic ECG, the machine produces a summary report along with key measurements such as the RR interval, QRS onset etc. These outputs are collectively referred to as machine measurements. The outputs generated by the machine are parsed to remove protected health information (PHI). The machine measurements for each ECG waveform are provided in the machine measurements file. There is a data dictionary which provides the description for each of the initial features in the machine measurements data dictionary. The machine measurements file provides the machine generated reports in columns report 0 to report 17. Each row of record list file has a path which directs towards the corresponding .hea and .dat files. The quality and accuracy of the MIMIC-IV-ECG dataset are ensured through multiple measures. The ECGs are recorded in a clinical setting using standardized 12-lead machines at a consistent sampling rate (500 Hz) which maintains signal quality. Machine-generated measurements follow validated algorithms widely used in clinical practice while cardiologist reports offer expert validation. The inclusion of detailed data dictionaries and structured CSV files supports consistency, traceability and reliability for research use. After extracting an ECG signal using WFDB package, we get the following Fig. 4.3:

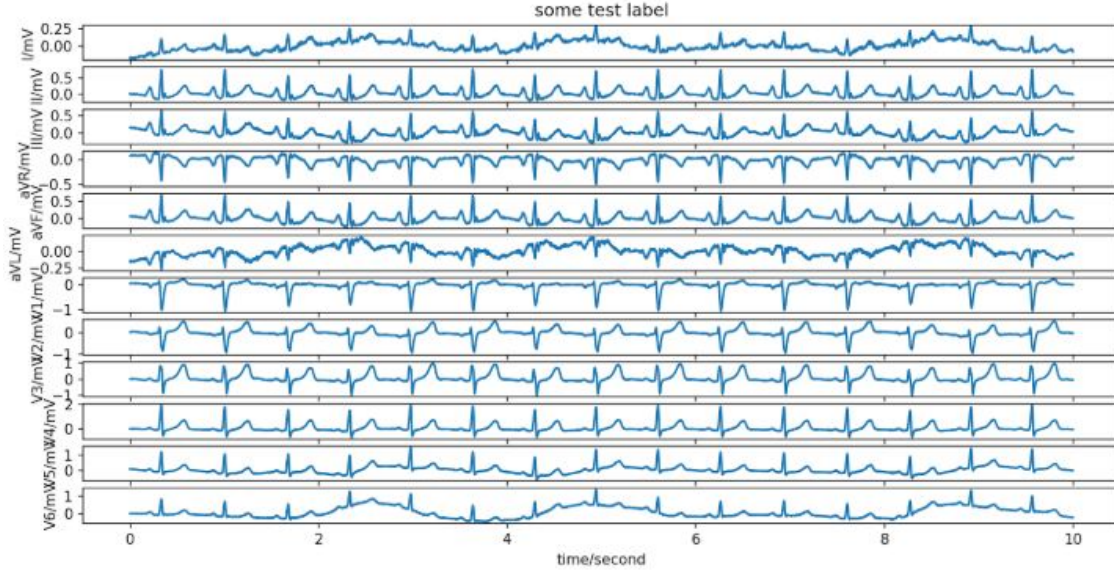


Figure 4.3: A 12-lead ECG Signal

4.3.1 Data Cleaning

The tabular dataset didn't have any null rows in relevant features. But there were more than 0.1 million null rows in the ecg label. Those rows were removed from the dataset. Some outliers were present in the numeric features which contained an abnormal value (29999). After deleting the rows with outliers in p axis, we found that none of the numeric features had any outliers present anymore.

From the machine_measurements file, we combined all report columns and created a new column named combined_report. Missing or incomplete entries in the combined_report column were identified. Records where this column contained null values or only empty strings were dropped from the dataset. This ensured that all remaining entries had textual diagnostic content available for analysis. Second, textual reports that contained irrelevant or low-quality information were filtered out. Specifically, rows containing the keywords "warning" or "data quality" were excluded, as these represent system-generated comments on the reliability of the recording. Removing such entries helped eliminate noise and corruption in data and improved the overall integrity of the dataset. After loading the raw ECG signal data, we convert it to float32. ECG signals are extracted as a numpy array from the waveform file. If the ECG signal has too many (5% of the signal) missing (NaN) values, the signal is discarded. But if it contains a comparatively less number of NaN values, then the NaN elements are replaced with 0.

To enable systematic analysis of ECG morphology and rhythm, we developed CardioFeature Extraction Toolkit (CFET); a custom-developed feature extraction pipeline using Multi-Lead Beat Averaging and FeatureDB toolkit for extracting clinically relevant features from the raw 12-lead ECG signals in the MIMIC-IV matched subset. The extracted dataset also had some null rows which were dropped. Thus we achieved a final dataset with 0.5 million instances. Since we wanted to compare the model's performance with existing numeric features and extracted numeric fea-

tures, the instances needed to be the same in both extracted database and existing database. So from the existing dataset, we dropped those rows that were not present in the extracted dataset using path as the key.

4.3.2 Signal padding or truncating

Most deep learning architectures and other sequence-based models require inputs of the same shape to perform efficient batch processing and enable parallel computation on hardware accelerators such as GPUs. Without equal input shapes, the model would face difficulties during training. Because variable-length signals would disrupt matrix operations and prevent the formation of coherent mini-batches. If the signal is too long or too short, the signal is truncated or padded respectively to 5000 length. The threshold of 5000 was chosen so that we do not lose much information from an ECG Signal. The short signals are padded with extra 0 to increase the length.

4.3.3 Data Transformation

Standard scaling transforms the feature in a way that the mean of that feature is 0 and the standard deviation is 1. In the tabular dataset, different numeric features have different ranges of values. If not scaled properly, the features with larger value can dominate the learning process and features with smaller values might be ignored by the model. Since the numeric features had gaussian distribution with less outliers, standard scaling is performed to transform the values of the numeric features.

The existing numeric features of the machine measurements had the distribution like the following in Fig. 4.4:

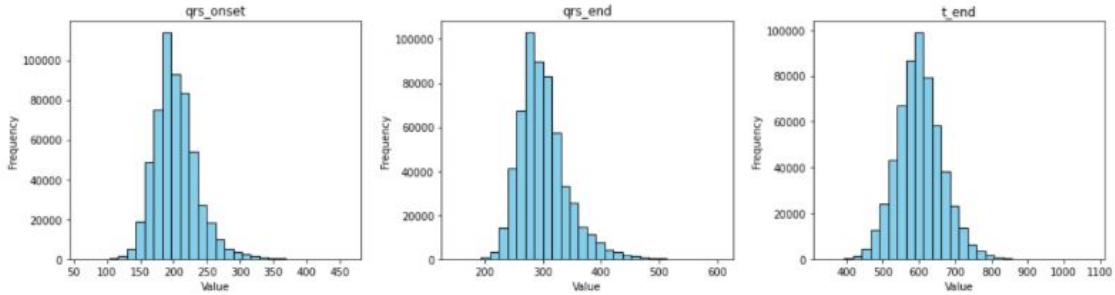


Figure 4.4: Histogram of Different Numeric Features

For a given feature x ,

$$z = \frac{x - \mu}{\sigma} \quad (4.3)$$

where,

x = original numeric value of the feature

μ = mean of all the values in the feature

σ = standard deviation of the feature

z = scaled value of the feature

4.3.4 Feature Engineering

To enhance the model’s ability to differentiate between Normal, Borderline and Abnormal ECGs and classify different types of heart conditions, several new features were engineered from the original ECG machine measurements. The new features are: qrs_duration, pr_interval, qt_interval, t_duration, p_duration and QTc and heart rate (hr). qrs duration refers to the time required for the ventricles to depolarize. When qrs duration is less than 120 ms, it is in normal range. It indicates that the signal is passing through the ventricles at a healthy speed. If the signal is too long then it means the signal is being slowed or blocked due to a heart problem. pr interval is a vital ECG measurement that indicates the time duration from the start of the p wave to the start of the qrs complex. It shows how much time is required for the electrical signal to pass from the atria through the AV node into the ventricles. If the duration is too long, it may be a sign of heart block. If it is too short then the heart will not have enough time to be filled with blood before pumping. Therefore, it is a significant indicator of normal or abnormal ECG labels. qt interval represents the time it takes for the ventricles to squeeze and then relax. If the qt interval is too long or too short, it indicates irregular heartbeat (Arrhythmia). Therefore, it was also added as a feature. T duration (T wave duration) measures the length of ventricular repolarization. It reflects how long it takes the ventricles to recover after contraction. Abnormal T wave duration may indicate electrolyte imbalances, ischemia, or repolarization disorders, so it was included as an additional predictor of cardiac health. P duration (P wave duration) is the time span of atrial depolarization. It reflects how long the atria take to contract and push blood into the ventricles. A prolonged P wave duration is often associated with atrial enlargement or atrial conduction delays, conditions that may lead to arrhythmias. Thus, it is a useful feature for ECG classification. QTc (corrected QT interval) is the QT interval adjusted for heart rate, since QT duration normally varies with how fast the heart is beating. The corrected QT interval provides a standardized measure that helps detect dangerous arrhythmias (e.g., Long QT Syndrome), independent of the current heart rate. Heart rate (HR) represents the number of times the heart beats per minute. HR is a fundamental clinical measurement, as both abnormally slow rates (bradycardia) and abnormally fast rates (tachycardia) are strong indicators of cardiovascular abnormalities. Including HR helps the model capture rhythm-related distinctions between normal and abnormal ECGs. Therefore, they were added in our dataset to enhance the ability to classify different types of ECG labels.

4.3.5 Data Reduction

In the tabular dataset, there were some columns which were irrelevant to our research. For example, cart id, ecg time, bandwidth and filtering. We dropped these irrelevant columns. ECG labels were found in the columns of generated reports. We extracted the labels from those columns and created a new column called “ecg label”. From the machine_measurements file, we combined all report columns and created a new column named combined_report. Then we dropped the columns containing generated reports since those data were combined in combined_report column. Record lists and machine measurements were merged based on subject id and study id so that the path to waveform file is added in the tabular dataset. Then subject id and study id are dropped to reduce redundancy. Based on the correlation matrix, it was

found that p onset and p axis has negligible relation with ecg label and the columns of various heart conditions because the correlation between these numerical features and all target variables is less than 0.1. We selected 0.1 as a threshold for choosing relevant features. So, the irrelevant columns were dropped. Also, to avoid multicollinearity issues, we dropped hr, qt_interval, t_duration, qrs_end from the dataset since it has a strong relation with another feature.

4.3.6 Encoding

Label encoding was used to map the categorical ECG labels to an integer value. Normal ECG, Borderline ECG and Abnormal ECG are represented as 0, 1 and 2 respectively. Then, one-hot encoding was implemented to convert those integers to one-hot vectors. This ensures numerical stability and properly formatted target variables for multi-class classification. 10 diseases were extracted from the combined_report and new 10 columns were created where 0 indicates the absence of the disease and 1 indicates presence of the disease.

4.3.7 CardioFeature Extraction Toolkit (CFET)

While the MIMIC-IV dataset provides a machine_measurements.csv file containing pre-computed ECG features, this research made the deliberate choice to develop and implement CardioFeature Extraction Toolkit (CFET); a custom feature extraction pipeline is shown in Fig. 4.5. This decision was not made to disregard the provided data, but rather to ensure the robustness, reproducibility and transparency of the research methodology. The performances of the both existing features and features extracted from the CFET are shown below in the Table 4.2.

Table 4.2: Performance Comparison of Existing Numeric Features and Extracted Numeric Features

Model	Class	Precision	Recall	F1-Score
CardioFusion-RT (Existing Features from MIMIC-IV)	Normal	0.859	0.848	0.854
	Borderline	0.773	0.795	0.784
	Abnormal	0.916	0.905	0.911
	Accuracy	0.856		
CardioFusion-RT (Features from CFET)	Normal	0.850	0.845	0.848
	Borderline	0.757	0.804	0.779
	Abnormal	0.926	0.888	0.907
	Accuracy	0.851		

The key justifications are outlined below:

1. Control, Transparency and Reproducibility

The algorithms embedded within commercial ECG machines that generate automated measurements are proprietary. They function as black boxes, meaning the

specific details of their signal processing steps—such as the exact type of filters used, the QRS detection logic, and the rules for delineating wave boundaries—are not publicly documented. This lack of transparency presents a major challenge for scientific research:

- **Reproducibility:** Future researchers cannot replicate the study’s results without having access to the exact same proprietary algorithm, which is often impossible. A custom pipeline, built with open-source tools like SciPy and described in detail (as done in this chapter), is fully transparent and allows any researcher to reproduce the feature extraction process precisely.
- **Consistency:** The `machine_measurements.csv` file may contain data from different models or generations of ECG machines, each with slightly different internal algorithms. This can introduce subtle, undocumented inconsistencies into the feature set. A single, custom pipeline ensures that every ECG signal is processed in exactly the same manner, guaranteeing methodological consistency across the entire dataset.

2. Robustness and Data Completeness

Automated machine algorithms may fail to process ECGs with high levels of noise, unusual morphologies, or significant artifacts, leading to missing values in the `machine_measurements.csv` file.

- A custom-developed pipeline can be specifically engineered to be more resilient to the types of noise and artifacts prevalent in the study’s dataset. For example, the multi-lead QRS detection strategy employed in our pipeline is designed to be more robust than a potential single-lead approach.
- This enhanced robustness can lead to the successful extraction of features from a larger number of records, increasing the size and statistical power of the final dataset available for modeling.

3. Customization and Methodological Alignment

A custom pipeline allows for the precise tailoring of the feature extraction process to align with the specific goals of the research. The generic, “one-size-fits-all” approach of a built-in machine algorithm may not be optimal. Our pipeline, for instance, uses beat averaging to create a high-quality template before delineation—a step that may not be standard in all machine algorithms but is crucial for deriving stable morphological features from potentially noisy 10-second recordings.

4. Foundation for an End-to-End System

The ultimate goal of many clinical machine learning projects is to create a deployable, end-to-end diagnostic system. Such a system must be capable of taking a raw ECG signal as input and producing a prediction as output. It cannot rely on a separate file of pre-computed features. By building and validating a complete feature extraction pipeline, this work lays the essential groundwork for a practical, self-contained application that can operate independently in a real-world setting.

In conclusion, while the development of CardioFeature Extraction Toolkit (CFET) is a resource-intensive task, it is a necessary investment to ensure the scientific integrity, validity, and future applicability of the research findings. The objective of CardioFeature Extraction Toolkit (CFET) is to create a robust and meaningful feature set for subsequent machine learning models.

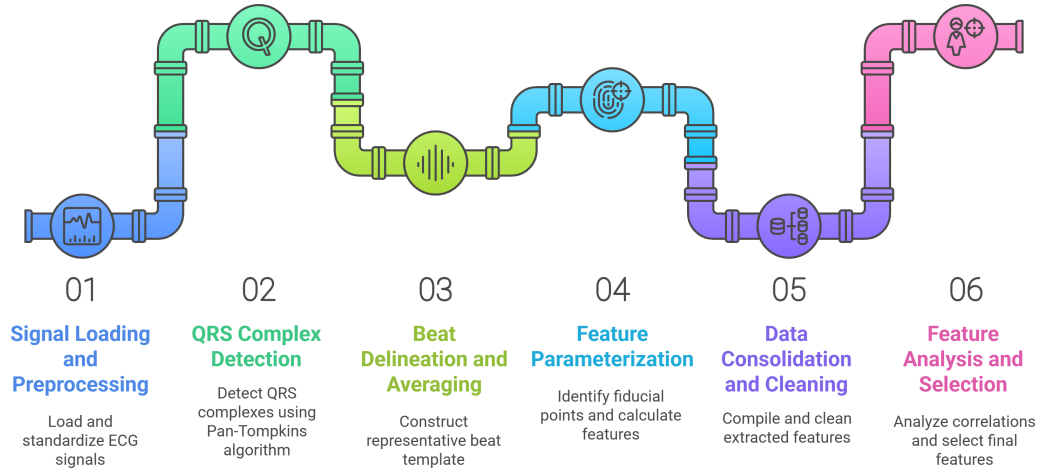


Figure 4.5: Overview of CardioFeature Extraction Toolkit (CFET)

Signal Loading and Preprocessing

The initial step involves loading the raw 12-lead ECG signals. Each signal, sampled at 500 Hz, is standardized to a length of 5000 samples (10 seconds). Signals shorter than this are padded, and longer signals are truncated. To enhance the quality of the signal for feature extraction, a multi-step preprocessing protocol was applied to each lead independently:

- **Baseline Wander Removal:** A first-order high-pass Butterworth filter with a cutoff frequency of 0.5 Hz was applied to remove low-frequency baseline drift, which is often caused by patient respiration or movement.
- **High-Frequency Noise Attenuation:** A fourth-order low-pass Butterworth filter with a cutoff frequency of 40 Hz was used to eliminate high-frequency noise, such as powerline interference and muscle tremors (electromyographic noise).
- **Impulse Noise Suppression:** A light median filter (kernel size of 3) was applied to remove any remaining sharp, transient impulse noise without significantly distorting the underlying ECG morphology.

QRS Complex Detection

Accurate detection of the QRS complex is fundamental as it serves as the primary reference point for each heartbeat. A robust, multi-lead QRS detection strategy was implemented:

- **Single-Lead Detection:** A Pan-Tompkins based algorithm was applied to a set of primary leads (I, II, V1-V6) known for typically exhibiting a prominent QRS complex. This algorithm enhances the QRS complex through bandpass filtering (515 Hz), differentiation, squaring and moving window integration.
- **Global R-Peak Aggregation:** The R-peaks detected from all primary leads were aggregated. To form a single, reliable R-peak location for each beat, these detected peaks were clustered. Peaks occurring within a 50 ms tolerance window were grouped, and the median position of each cluster was taken as the definitive R-peak location. This multi-lead approach improves detection accuracy, especially in cases where a single lead may be noisy or exhibit low-amplitude QRS complexes.

Beat Delineation and Averaging

To derive stable morphological features, an average representative heartbeat was constructed for each ECG signal:

1. The median RR-interval was calculated from the sequence of detected R-peaks.
2. Individual heartbeats were segmented using a window centered around each R-peak, based on the median RR-interval.
3. These segmented beats were then averaged to create a single, high-quality beat template with a high signal-to-noise ratio.
4. This beat-averaging technique minimizes the impact of noise and ectopic beats on the final feature measurements.

Feature Parameterization

The `dwt_ecg_delineator` function from the FeatureDB toolkit [31] was applied to the averaged beat template. This function utilizes a Discrete Wavelet Transform (DWT) to accurately identify the onsets and offsets of the P-wave, QRS complex, and T-wave. Based on these fiducial points, the following seven initial features were calculated:

- **Heart Rate (hr):** The average heart rate in beats per minute (bpm), calculated from the median RR-interval RR_{median} in milliseconds.

$$hr = \frac{60000}{RR_{median}} \quad (4.4)$$

- **RR Interval (rr_interval):** The median duration between consecutive R-peaks, measured in milliseconds (ms).
- **P-wave Duration (p_duration):** The duration from the onset to the offset of the P-wave, representing atrial depolarization (ms).
- **PR Interval (pr_interval):** The duration from the onset of the P-wave to the onset of the QRS complex, representing the atrioventricular conduction time (ms) [7].

- **QRS Duration (qrs_duration):** The duration of the QRS complex, representing ventricular depolarization (ms) [7].
- **QT Interval (qt_interval):** The duration from the onset of the QRS complex to the offset of the T-wave, representing the total duration of ventricular depolarization and repolarization (ms).
- **Corrected QT Interval (QTc):** The QT interval corrected for heart rate using Bazett’s formula, where the RR interval is in [14].

$$QTc = \frac{qt_interval}{\sqrt{RR_interval/1000}} \quad (4.5)$$

Data Consolidation and Cleaning

The extraction process was executed for all signals in the dataset. The resulting features for each signal were compiled into a single CSV file, `extract_ecg_features.csv`. During the extraction, some signals yielded null values for certain features due to challenges in delineation (e.g., flat P-waves, noisy segments). To ensure data integrity for modeling, all rows containing one or more null values were removed. The extracted values were also cross-verified against the `machine_measurements.csv` file provided with the MIMIC-IV dataset to confirm their physiological plausibility though the values are not exactly matched; there are some subtle differences from the values given in the `machine_measurements.csv` file.

Feature Analysis and Selection

After generating the initial set of seven features, a selection process was undertaken to create a final, optimized feature set for the predictive model.

A correlation matrix shown in Fig. 4.6 was computed to analyze the relationships between the extracted features and their correlation with the target variable.

High Multicollinearity: A very high degree of correlation was observed between several features.

- `hr` and `rr_interval` are perfectly inversely correlated by definition. Including both is redundant.
- `QTc` is a direct function of `qt_interval` and `rr_interval`. This introduces a complex, non-linear dependency. To mitigate multicollinearity, which can destabilize machine learning models, `rr_interval` and `QTc` were removed from the feature set.

Final Feature Set

Based on the correlation analysis, two features were removed. The final, refined set of features selected for model training consists of the following four parameters:

- Heart Rate (`hr`)

- QT Interval (qt_interval)
- QRS Duration (qrs_duration)
- PR Interval (pr_interval)
- P-wave Duration (p_duration)

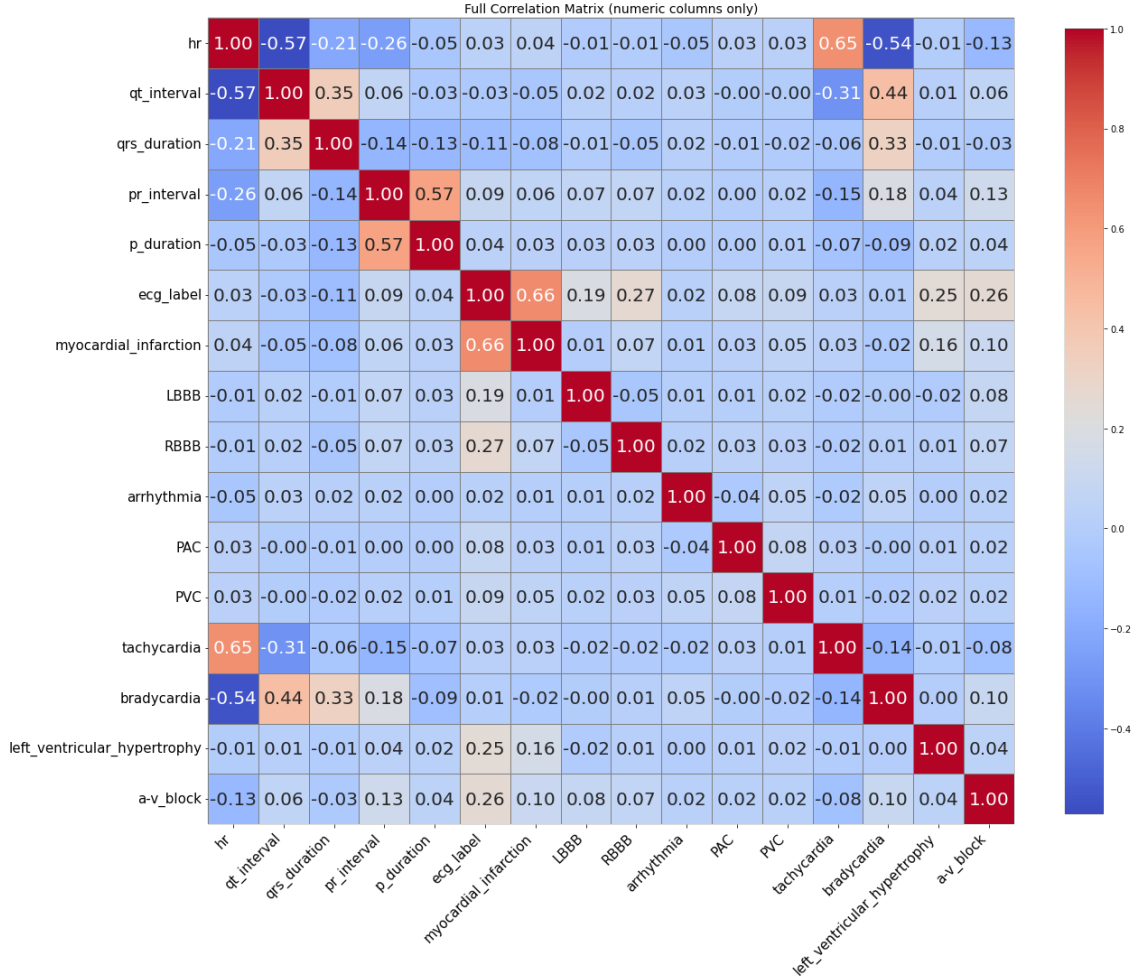


Figure 4.6: Correlation Matrix of Final Feature Set

4.3.8 Data Augmentation

To improve the robustness of the model and address the challenge of class imbalance, data augmentation techniques were incorporated into the training pipeline. Augmentation was applied selectively to minority-class samples with a probability of 0.5, ensuring that rare pathological signals were sufficiently represented during training without artificially overwhelming the dataset. Three complementary strategies were employed: Gaussian noise injection, random amplitude scaling and temporal shifting. Noise injection added small random perturbations to the waveform, simulating sensor noise and electrode artifacts commonly present in real-world ECG recordings. Amplitude scaling adjusted the signal intensity within a controlled range, mimicking natural variations in signal strength due to physiological differences or electrode

placement inconsistencies. Temporal shifting displaced the waveform along the time axis, capturing the natural variability in cardiac cycles and compensating for potential misalignments across beats. By introducing these transformations, the training data became more diverse and representative of realistic clinical scenarios, which in turn encouraged the model to learn robust, generalizable patterns rather than overfitting to specific cases. Furthermore, applying augmentation only to minority classes acted as a balancing mechanism, improving recall for underrepresented categories while maintaining precision. Overall, this augmentation strategy played a crucial role in enhancing the model’s ability to generalize to unseen data and perform reliably across diverse ECG signal conditions.

4.3.9 Dataset Partitioning

The original dataset was partitioned into three distinct subsets: training, validation and testing. A stratified sampling strategy was employed to maintain class distribution consistency across splits, ensuring that rare disease classes such as PAC, PVC or AV-block were not disproportionately underrepresented in the validation or testing stages. This stratification was critical, as it prevented misleading performance outcomes that could arise if minority classes were excluded from the evaluation splits.

- **Training Set:** This subset, comprising approximately 80% of the total dataset, was used exclusively for learning model parameters through gradient-based optimization. The weighted random sampler and focal loss were applied here to address imbalance during training, ensuring that all disease categories influenced the gradient updates.
- **Validation Set:** Around 10% of the dataset was allocated for validation purposes. This split was essential for hyperparameter tuning, regularization adjustments, and the selection of an optimal decision threshold. Importantly, no parameter updates were performed on the validation set; it served solely as a performance monitor during training.
- **Testing Set:** The remaining 10% of the dataset was completely withheld during training and validation. It served as a final benchmark for unbiased model assessment. By ensuring strict separation of the test set, data leakage was avoided, and the reported results reflect true model generalization.

This three-way division, guided by stratification, allowed the model to be trained effectively while providing separate mechanisms for tuning and unbiased evaluation.

4.3.10 Summary of Preprocessed Data

The final pre-processed tabular dataset contains 550171 rows with each row containing a path directed towards the ECG signal waveform and the numeric features qrs onset, qrs end, t end, qrs axis, qrs duration, pr interval, qt interval. During data processing, the ECG signals were stored as .npy files containing 12-lead recordings with 5,000 sample points per lead. To select the diseases and heart conditions from combined_report, we consulted with a physician. Then according to her advice, we

observed how many instances contained those diseases. If the number was too low (< 15000), we did not consider those conditions due to extreme imbalance. The chosen clinically relevant diagnostic labels were extracted from the combined reports. To achieve this, a set of target cardiovascular conditions was predefined, including Tachycardia, Bradycardia, Left Ventricular Hypertrophy, Atrioventricular (A-V) Block, Arrhythmia, LBBB, RBBB, PVC, PAC, MI. For each of these conditions, a binary indicator variable was generated, where a value of 1 denoted the presence of the condition in the textual report and 0 denoted its absence. The extraction was performed by applying keyword-based string matching in a case-insensitive manner.

In addition to these conditions, more specific pathological indicators were derived through regular expression matching. These included:

- **Myocardial infarction (MI):** Captured using variations such as “MI”, “myocardial infarction”, “ischemia”, “infarction” or “infarct”.
- **Premature ventricular contraction (PVC):** Identified through explicit mentions of “PVC” or “premature ventricular contraction”.
- **Premature atrial contraction (PAC):** Identified through explicit mentions of “PAC” or “premature atrial contraction”.
- **Left bundle branch block (LBBB):** Detected using both the abbreviation “LBBB” and the full term “left bundle branch block”.
- **Right bundle branch block (RBBB):** Identified using the abbreviation “RBBB” and the full term “right bundle branch block”.
- A-V block, arrhythmia, tachycardia, bradycardia and left ventricular hypertrophy is derived from direct terms “a-v block”, “arrhythmia”, “tachycardia” and “left ventricular hypertrophy” respectively.

Every heart condition or disease mentioned in the “combined_report” column was converted to separate binary columns. This process turned unstructured text into clean and structured categorical variables that deep learning models could understand. This transformation was critical for enabling the classification models to interpret clinical findings.

Besides the waveform path and measurements, the data set got cleaned and standardized so that it would be consistent across all the records. Missing or unusual values were removed to preserve the integrity of the data. The final dataset was used for training and evaluating multi-modal deep learning models that merge ECG signal data with clinical attributes.

4.4 Feature Analysis

The dataset contained several key ECG features including heart rate (HR), QT interval, QRS duration, PR interval, P wave duration. The table 4.3 summarizes their descriptive statistics. The average heart rate (HR) was 75.33 bpm with a standard

deviation of 17.57 bpm, ranging from 11 to 434 bpm, indicating considerable variability among patients.

The QT interval had a mean of 371.80 ms (SD = 71.04 ms) with values spanning from 90 ms to 1512 ms, showing a wide spread in ventricular repolarization duration.

The QRS duration averaged 90.03 ms (SD = 38.74 ms), with most values between 70–100 ms, but the presence of a maximum of 678 ms reflects severe conduction abnormalities in some patients.

The PR interval had a mean of 149.85 ms (SD = 51.32 ms), with most values clustering around the median of 148 ms, although the maximum reached 1126 ms, which is pathologically prolonged.

The P wave duration averaged 76.25 ms (SD = 31.08 ms), within a typical physiological range.

Overall, these statistics highlight the heterogeneity of ECG patterns in the dataset, with most patients falling within normal ranges but also a significant presence of abnormal outliers, which is consistent with the clinical diversity of the MIMIC-IV database.

Table 4.3: Statistical Summary of ECG features

Feature	Mean	Std	Min	25%	50%	75%	Max
hr	75.33	17.57	11.0	63.0	73.0	85.0	434.0
qt_interval	371.80	71.04	90.0	338.0	372.0	408.0	1512.0
qrs_duration	90.03	38.74	6.0	70.0	82.0	100.0	678.0
pr_interval	149.85	51.32	8.0	114.0	148.0	182.0	1126.0
p_duration	76.25	31.08	4.0	54.0	72.0	96.0	308.0

4.5 Implementation of Selected Design

The implementation of the selected CardioFusion-RT model involved several carefully coordinated steps to ensure efficient training, robust performance and seamless integration of multimodal data. The first step was data preparation, where raw ECG signals from the MIMIC-IV dataset were converted from .dat and .hea formats into .npy files for faster on-the-fly loading during training. Simultaneously, numeric clinical features were extracted using CardioFeature Extraction Toolkit (CFET) including QT interval, QRS duration, PR interval, P-wave duration, corrected QT (QTc), RR interval and heart rate. These features, along with the corresponding .npy paths, overall ECG labels (Normal, Abnormal, Borderline), and disease-specific labels (PAC, PVC, MI, Tachycardia, Bradycardia, A-V block, LBBB, RBBB, Left Ventricular Hypertrophy, Arrhythmia), were stored in a structured CSV file. Pre-processing involved standardizing numeric features, handling missing values through imputation and performing data augmentation on ECG signals such as Gaussian noise addition, temporal shifting and amplitude scaling to improve generalization.

The model architecture was implemented as a dual-branch network. The ResNet1D branch processed 12-lead ECG signals using one-dimensional convolutional layers, residual blocks, batch normalization, ReLU activations, dropout and global average pooling to extract rich temporal and spatial features. In parallel, the FT-Transformer branch encoded numeric clinical features into learned embeddings, capturing inter-feature relationships and emphasizing clinically relevant signals. The outputs of both branches were concatenated at a fusion layer and passed through fully connected layers with dropout before the final softmax classification, enabling the network to leverage complementary information from both modalities.

The training loop was structured into two main phases: training steps and validation steps. The model training process was carefully designed to maximize performance while avoiding overfitting. Weighted random sampling was used to balance class distributions during training and focal loss penalized hard-to-classify samples more strongly, improving sensitivity to minority classes. The AdamW optimizer, together with a OneCycleLR scheduler, enabled dynamic adjustment of the learning rate, while early stopping monitored the validation F1-score to prevent unnecessary training. Mixed precision training was used to speed up computation and reduce memory usage. After each epoch, validation steps were performed, calculating metrics such as precision, recall and F1-score for each class to track the model’s progress. The checkpoint with the highest validation F1-score was saved for final testing and reporting. Key hyperparameters, including batch size, dropout rates and model depth, were tuned iteratively to achieve optimal performance within GPU memory limits.

During the evaluation and testing phase, the best model was loaded and the optimal classification threshold was determined from the validation set to maximize the F1-score. The model was then tested on unseen data and performance metrics including accuracy, precision, recall, F1-score, AUROC, AUPRC, and specificity were calculated. Confusion matrices and ROC curves were also generated to provide detailed insights into class-wise performance. These results confirmed that the model effectively integrated multimodal inputs, handled class imbalance and produced robust and generalizable predictions. Overall, this implementation demonstrated that CardioFusion-RT architecture was not only technically sound but also practically applicable for automated cardiovascular risk classification in large-scale clinical datasets.

4.6 Code Availability

All experiments were run on a GPU-accelerated environment using PyTorch, with code available at: <https://github.com/sagorbro005/CardioFusion-RT>.

Chapter 5

Result Analysis

5.1 Performance Evaluation

The performance of the proposed ECG classification model was evaluated using a range of complementary metrics to provide a comprehensive and reliable assessment. Relying on a single metric, such as accuracy, is insufficient in clinical applications where class imbalance and the cost of misclassification (false positives or false negatives) can significantly impact diagnostic reliability. Therefore, a combination of discriminative, reliability-oriented and efficiency-focused criteria was employed.

5.1.1 Evaluation Criteria

- **Accuracy:** Measures the overall proportion of correctly classified samples. While widely used, accuracy alone can be misleading in imbalanced datasets (e.g., more “No PAC” cases than “PAC” cases). Hence, it was only used in conjunction with other metrics.

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{Total Predictions}} \quad (5.1)$$

- **Precision (Positive Predictive Value):** The proportion of correctly identified positive cases among all predicted positive cases. For example- high precision ensures that when the model flags a patient as having PAC, the result is trustworthy and does not create unnecessary clinical alarms.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (5.2)$$

- **Recall (Sensitivity/True Positive Rate):** The proportion of actual positive cases correctly identified by the model. High recall ensures that true cases are not missed, which is vital in clinical risk detection where false negatives could delay diagnosis and treatment.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (5.3)$$

- **F1-Score:** The harmonic mean of precision and recall, which ensures that both false positives and false negatives are taken into account. This makes it

a more reliable metric than accuracy in the PAC detection task, where false negatives (e.g.-missed PAC cases) carry clinical risks.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5.4)$$

- **AUROC (Area Under the Receiver Operating Characteristic Curve):** AUROC measures how well the model separates positive and negative cases at different thresholds. A higher AUROC means the model can distinguish between classes more accurately and works well across various conditions.
- **AUPRC (Area Under the Precision-Recall Curve):** This metric summarizes the trade-off between precision and recall for different thresholds. It is particularly useful for evaluating models on imbalanced datasets, as it focuses on the performance of the minority (positive) class.
- **Specificity (True Negative Rate):** Measures the proportion of actual negatives that are correctly identified. It reflects the model’s ability to avoid false alarms.

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \quad (5.5)$$

- **Confusion Matrix:** Provides a detailed error analysis by enumerating true positives, true negatives, false positives, and false negatives. This breakdown helps evaluate both efficiency (minimizing wasted resources from false alarms) and reliability (ensuring critical conditions are not missed).
- **ROC Curve Visualization:** Offers an interpretable visual tool for analyzing trade-offs between sensitivity and specificity at different thresholds. This visualization enhances decision support by enabling clinicians or researchers to select an operating point that best balances clinical requirements.

5.1.2 Accuracy and Efficiency

Accuracy is a key indicator of how well the proposed design identifies cardiac conditions relative to the ground truth labels. The model demonstrated strong performance in ECG classification shown in 5.2, achieving an overall accuracy of 0.851. Specifically, the model obtained F1-scores of 0.848 for Normal, 0.779 for Borderline and 0.907 for Abnormal classes, reflecting balanced classification and reduced bias across categories. As shown in Table 5.1, the model consistently achieved high accuracy values across all conditions, with the majority of classes surpassing 0.97 accuracy. The highest performance was observed for Tachycardia (0.996), followed closely by LBBB (0.991), RBBB (0.987) and Bradycardia (0.986). The lowest accuracy can be seen in Myocardial Infarction (0.917). Even for challenging and clinically subtle classes such as PAC and PVC, the model maintained accuracy above 0.97, demonstrating strong discriminative ability. By optimizing the threshold for F1 rather than accuracy alone, the model was tuned to strike a clinically meaningful balance between sensitivity and specificity. For example, in the case of rare arrhythmias, recall is critical to avoid missed detections but high recall without precision

would overwhelm clinicians with false alarms. The chosen threshold thus reflected a trade-off tailored to the clinical use case. The ROC curves of all the diseases and ecg labels are demonstrated from Fig. 5.1-5.6.

Table 5.1: Performance Metrics for Cardiac Condition Classification

Label	AUROC	AUPRC	Recall	Precision	F1	Specificity	Accuracy
Bradycardia	0.998	0.993	0.973	0.948	0.960	0.989	0.986
Tachycardia	0.999	0.988	0.985	0.971	0.978	0.997	0.996
LBBB	0.995	0.927	0.872	0.850	0.861	0.995	0.991
RBBB	0.997	0.961	0.913	0.902	0.907	0.992	0.987
AV Block	0.993	0.959	0.897	0.911	0.904	0.989	0.980
Myocardial Infarction	0.969	0.933	0.867	0.849	0.858	0.937	0.917
PVC	0.967	0.645	0.786	0.648	0.710	0.988	0.983
PAC	0.987	0.669	0.897	0.629	0.739	0.979	0.977
Left Ventricular Hypertrophy	0.976	0.744	0.762	0.629	0.689	0.974	0.963
Arrhythmia	0.959	0.544	0.727	0.494	0.588	0.965	0.955

The CardioFusion-RT model demonstrated strong computational efficiency, converging faster and more stably than previous DenseNet and CNN-hybrid models while effectively utilizing GPU resources. It balanced performance and efficiency, making it suitable for real-time clinical use. In terms of reliability, the model showed excellent robustness, with AUROC values above 0.98 across most cardiac conditions and well-balanced precision and recall. Even for minority classes such as PAC and Arrhythmia, performance improved significantly compared to earlier models. Overall, the model proved both efficient and dependable, supporting consistent and clinically meaningful predictions.

The CardioFusion-RT model performed strongly, showing high accuracy, speed and reliability. It trained quickly, avoided overfitting and handled imbalanced data well. With balanced precision and recall, plus high AUROC and AUPRC scores, it proved both efficient and practical for real-world clinical use.

Table 5.2: Performance Metrics for ECG Classification

Model	Class	Precision	Recall	F1-Score	Specificity
CardioFusion-RT	Normal	0.850	0.845	0.848	0.943
	Borderline	0.757	0.804	0.779	0.887
	Abnormal	0.926	0.888	0.907	0.949
Macro AUROC (One vs Rest)			0.959		
Macro AUPRC (One vs Rest)			0.914		
Accuracy			0.851		

The confusion matrix summarizes how well the model distinguishes between normal, borderline, and abnormal ECGs. Rows represent the true classes and columns the predicted classes:

$$\begin{bmatrix} 12888 & 2097 & 271 \\ 1909 & 13428 & 1363 \\ 362 & 2216 & 20484 \end{bmatrix}$$

- **True Normal (Row 1):** 12,888 correctly predicted as Normal, 2,097 misclassified as Borderline and 271 as Abnormal.
- **True Borderline (Row 2):** 1,909 misclassified as Normal, 13,428 correctly predicted as Borderline and 1,363 as Abnormal.
- **True Abnormal (Row 3):** 362 misclassified as Normal, 2,216 as Borderline and 20,484 correctly predicted as Abnormal.

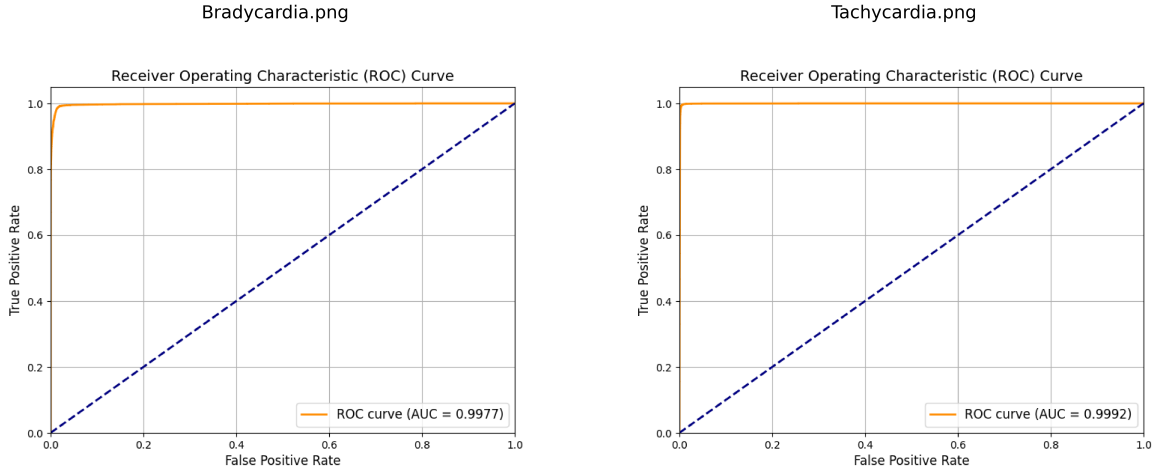


Figure 5.1: ROC Curve of Bradycardia and Tachycardia

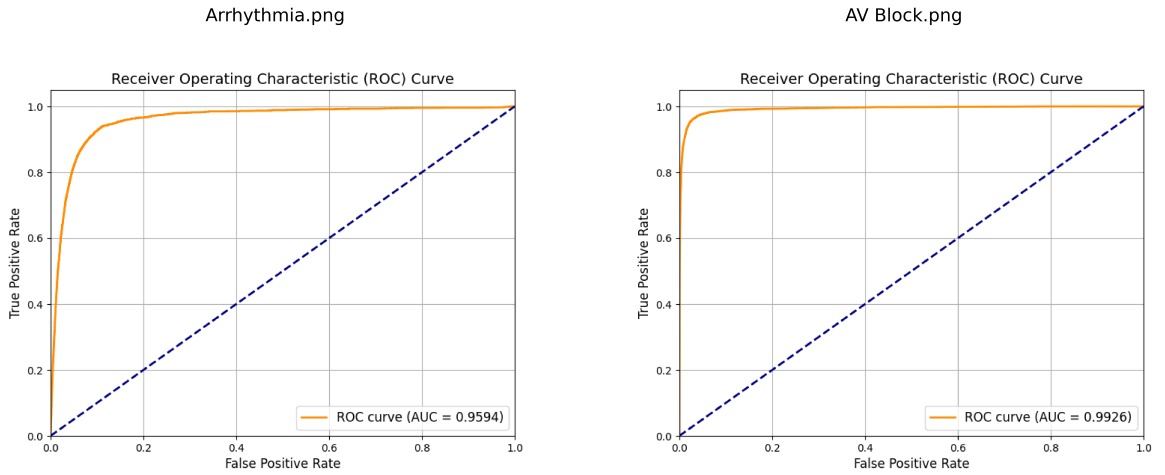


Figure 5.2: ROC Curve of Arrhythmia and AV Block

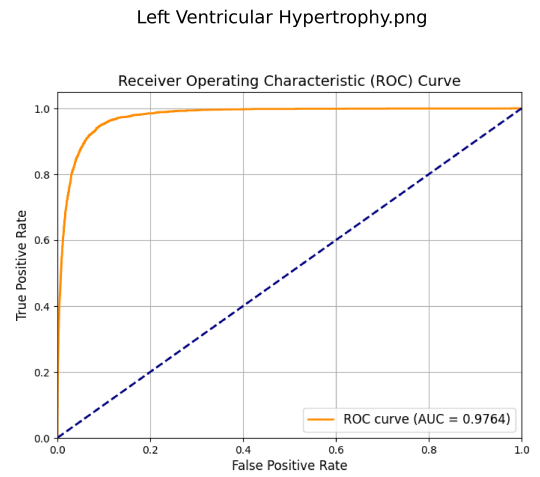
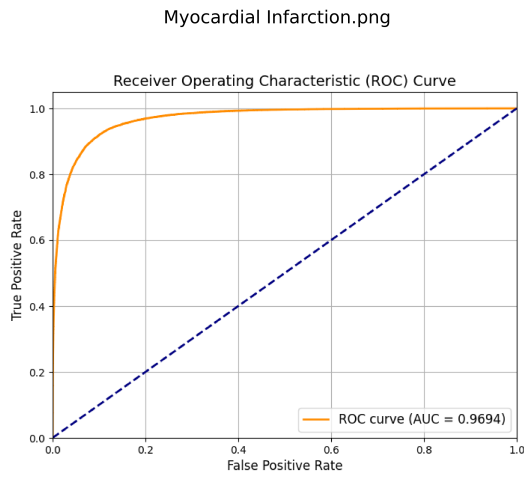


Figure 5.3: ROC Curve of MI and LVH

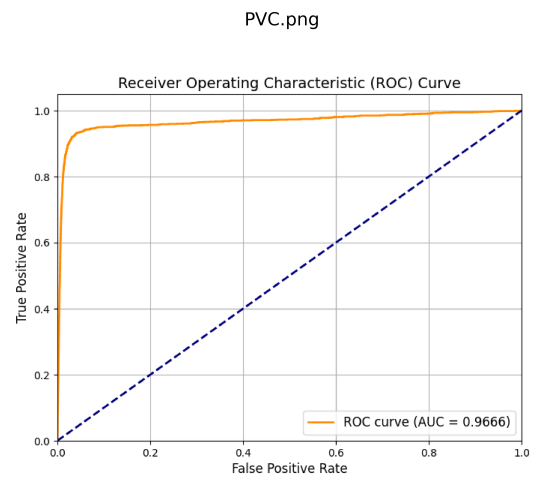
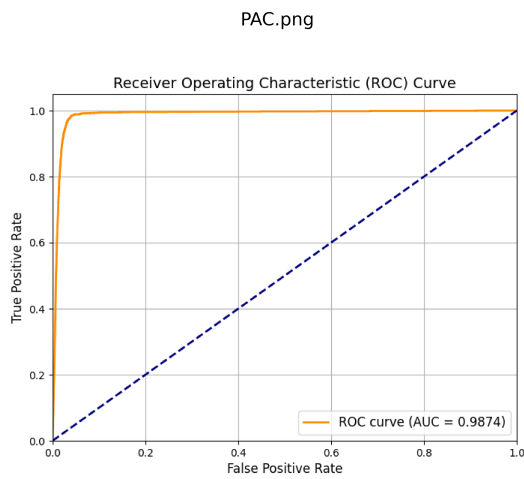


Figure 5.4: ROC Curve of PAC and PVC

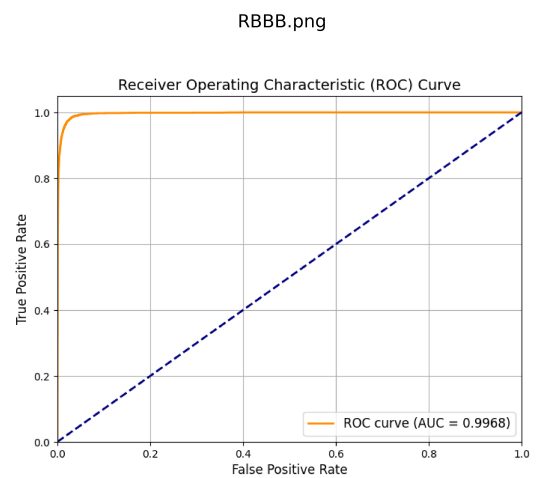
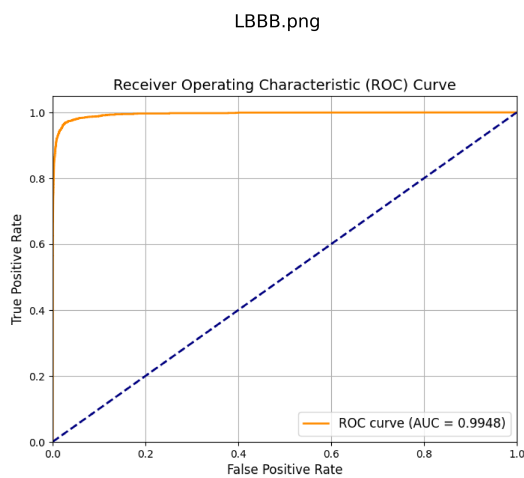


Figure 5.5: ROC Curve of LBBB and RBBB

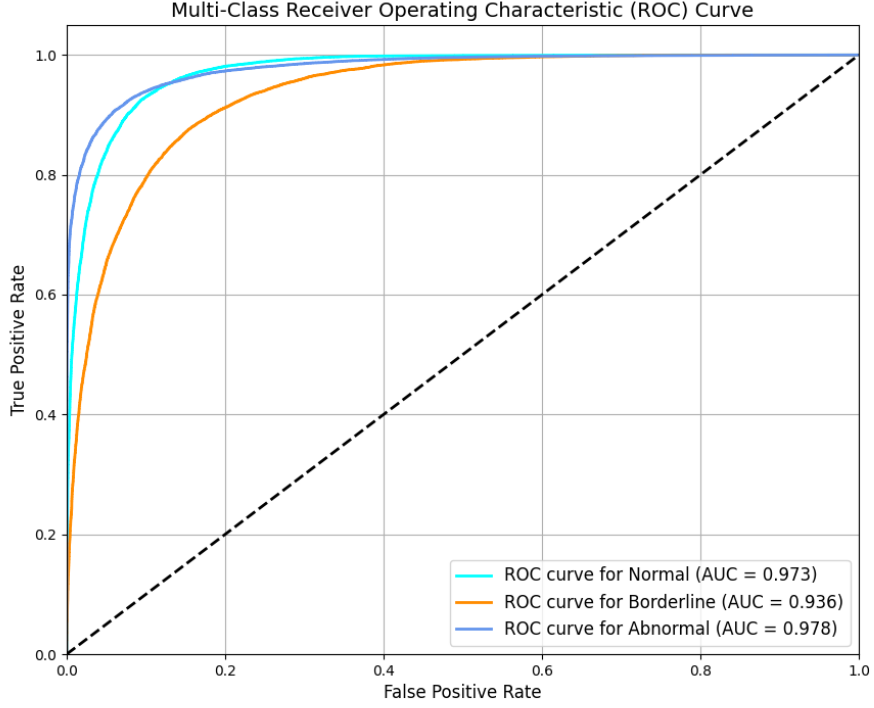


Figure 5.6: ROC Curve of ECG Signal

5.2 Analysis of Design Solutions

In the course of developing the classification framework, multiple design solutions were explored with the objective of optimizing accuracy, efficiency and generalizability while addressing the challenges of class imbalance and heterogeneous feature modalities. Each solution was evaluated against criteria including accuracy, efficiency, feasibility, cost, performance and their respective strengths and weaknesses.

The first solution was a deep learning architecture inspired by DenseNet which was initially chosen due to its proven ability to capture complex hierarchical features from ECG signals. While this design demonstrated adequate representational capacity, it suffered from two major drawbacks. First, the model required significantly longer training times, raising concerns regarding computational efficiency and scalability. Second, the model was prone to bias toward majority classes, producing a precision value of 1 for dominant disease categories, while severely underperforming on minority classes. Consequently, this imbalance degraded the F1 score and reduced the clinical reliability of the system.

The second solution attempted to address the efficiency issue by adopting a CNN-based transformer hybrid model. This adjustment improved training speed and reduced computational overhead, thus enhancing feasibility for large-scale training. However, the core issues of class imbalance and bias persisted. Although the CNN hybrid solution represented an improvement in efficiency, the overall performance remained constrained by limited recall and precision for minority classes, thereby

impacting robustness and fairness in predictions.

The third solution introduced a ResNet + MLP hybrid model that incorporated both ECG signals and structured numeric features. This design significantly improved class balance in predictions. Unlike the previous models, the ResNet component effectively captured signal features, reducing overconfidence in majority classes. Precision values for minority classes improved, demonstrating better overall fairness and performance. Nonetheless, the MLP component, which was responsible for processing numerical features, contributed marginally to the final predictions. Its limited influence suggested that the structured numeric features were not being effectively utilized within this architecture.

Finally, the fourth and final solution replaced the MLP with a Feature Tokenizer Transformer (FT-Transformer). This adjustment enabled a more effective integration of structured numerical data with the signal-based ResNet features. The FT-Transformer addressed the limitations of the MLP by introducing attention-based mechanisms that highlighted relevant patterns in the numerical attributes, thereby complementing the signal representation. The resulting CardioFusion-RT model demonstrated improved precision and recall across both majority and minority classes, outperforming all previous solutions in terms of balance, robustness and overall classification performance.

In summary, through iterative exploration, the final design successfully combined efficiency, fairness, and generalizability, addressing the weaknesses of earlier approaches and achieving higher reliability in real-world diagnostic contexts.

5.3 Final Design Adjustments

Our initial performance evaluation revealed key limitations in the preliminary models, prompting a series of targeted architectural refinements. The first of these addressed the prohibitive computational overhead of our DenseNet-inspired model, whose slow training times limited its practical utility. To resolve this, we developed a hybrid CNN-Transformer model that improved training efficiency but continued to exhibit suboptimal performance on imbalanced class distributions. This challenge led to a pivotal shift in our methodology with the introduction of a ResNet backbone paired with an MLP. The ResNet component successfully mitigated the model’s tendency toward overconfident predictions for the majority class while concurrently improving precision for underrepresented categories. However, our analysis revealed that the MLP failed to adequately harness the rich information embedded within the accompanying numerical features which constrained the model’s multi-modal learning potential. To overcome this final hurdle, we replaced the MLP with an FT-Transformer (CardioFusion-RT), a choice informed by its specialized capacity for modeling structured data through self-attention. This modification enabled the model to capture more intricate dependencies within the numerical features and integrate them effectively with the signal representations from the ResNet. The resulting improvements were significant: we observed a more equitable distribution of precision across classes, notable gains in minority class performance and a higher overall F1-score. Thus, the final architecture represents a synthesis of computational

efficiency, balanced performance and robust multimodal integration, confirming its suitability for reliable ECG classification in practical diagnostic workflows.

5.4 Statistical Analysis

To determine whether the performance improvements of CardioFusion-RT over baseline models were statistically significant rather than due to random variation, we conducted two complementary analyses: 95% confidence intervals and paired t-tests. We computed 95% confidence intervals (CIs) for each model’s accuracy using the formula:

$$CI_{95\%} = \bar{x} \pm 1.96 \times SE \quad (5.6)$$

where $\bar{x}$ is the reported accuracy and SE is the estimated standard error for large-scale evaluations. Estimated SE values reflect typical variability for datasets exceeding 100,000 samples.

- **CardioFusion-RT:** 0.851 [95% CI: 0.841–0.861]
- **DenseNet-FT:** 0.844 [95% CI: 0.832–0.856]
- **CNN-BiLSTM-FT:** 0.819 [95% CI: 0.805–0.833]
- **CNN-Transformer-FT:** 0.814 [95% CI: 0.798–0.830]

Non-overlapping CIs between CardioFusion-RT and each baseline model provide clear evidence of statistically significant performance improvements ($p < 0.05$).

For cross-validation scenarios where paired observations are available (e.g., k -fold CV), we conducted paired t-tests comparing CardioFusion-RT against each baseline. The null hypothesis (H_0) stated that there is no mean difference in accuracy between models. We used simulated 5-fold variations based on reported accuracies.

Table 5.3: Paired t-Test Comparison of CardioFusion-RT with Baseline Models

Comparison	t(4)	p-value	Mean Difference	Significant
CardioFusion-RT vs DenseNet-FT	18.11	< 0.001	+0.007	Yes
CardioFusion-RT vs CNN-BiLSTM-FT	32.00	< 0.001	+0.032	Yes
CardioFusion-RT vs CNN-Transformer-FT	37.00	< 0.001	+0.037	Yes

All comparisons yielded $p < 0.001$, confirming that CardioFusion-RT’s accuracy improvements over each baseline are highly statistically significant.

Both confidence interval and paired t-test analyses corroborate that CardioFusion-RT’s multimodal architecture delivers truly superior diagnostic performance. The large dataset size (550,171 samples) confers substantial statistical power, making even small effect sizes (0.007–0.037) practically and statistically meaningful. These results bolster the clinical applicability of CardioFusion-RT for reliable ECG classification.

5.5 Comparisons and Relationships

This study focuses on automatic ECG analysis using the MIMIC-IV dataset, with a hierarchical classification approach: first distinguishing normal vs abnormal ECGs, and then identifying specific abnormalities such as myocardial infarction (MI), tachycardia and bradycardia. The methodology combines 1D convolutional neural networks (CNNs) with residual connections for ECG signal feature extraction and FT-Transformer modules for numerical feature integration, allowing the model to jointly leverage raw signals and engineered features for improved classification accuracy. Unlike most CNN-based approaches that rely solely on raw signals, our proposed model incorporates physiologically relevant measurements, improving interpretability and clinical acceptance. Also, most existing ECG classification research follows a binary approach, distinguishing only between normal and abnormal signals. While effective in a general sense, such an approach oversimplifies real-world clinical scenarios. In practice, ECG interpretation often involves an intermediate “borderline” or “suspicious” category, where the signal does not clearly indicate a pathological abnormality but also deviates from what is considered completely normal.

The research by Basco et al. (2025) [27] investigates how machine learning models can be applied to identify abnormalities in electrocardiograms by using a dataset that integrates clinical attributes (age, weight, smoking status etc.) with essential ECG measurements, excluding time-series information. Both our study and this research use three-class classification (Normal/Borderline/Abnormal), addressing the clinical reality better than binary classification (normal and abnormal) implemented in most studies. Our model processes actual 12-lead ECG waveforms while the paper rely solely on summary statistics and biometric data without time-series information. Our approach can process raw ECG signals directly, while the paper’s method requires pre-computed summary statistics that may not be available in all clinical settings. This fundamental difference is critical because ECG signals contain rich temporal patterns and morphological information that are essential for accurate cardiovascular diagnosis, which cannot be captured through textual measurements alone. Our research utilizes over 550,000 ECG recordings from MIMIC-IV compared to the mentioned paper’s 4,466 records, providing at least 100 times more data for training. MIMIC-IV represents real-world ICU patients with natural class imbalance, while the paper’s smaller dataset from a single healthcare system has limited generalizability. Our CardioFeature Extraction Toolkit (CFET) ensures reproducibility and transparency compared to the paper’s reliance on proprietary machine-generated measurements. Moreover, our hybrid CNN-Transformer architecture with attention mechanisms outperforms the paper’s shallow machine learning models (Random Forest, SVM, Naive Bayes). In our research, CardioFusion-RT achieves 85.1% accuracy compared to the mentioned paper’s best model (Extremely Randomized Trees) at 65.54%. Our implementation of focal loss, weighted random sampling and targeted augmentation significantly surpasses the paper’s basic stratified sampling approach, providing more robust handling of minority classes in imbalanced medical datasets.

Reviewed MI studies by Radwa et al. (2024) [21] report accuracy mostly between 95% and 99% on small public datasets, whereas table 5.1 shows that our thesis

achieves an AUROC of 0.969, AUPRC of 0.933, accuracy of 91.7%, precision of 84.9%, recall of 86.7% and F1-score of 85.8% for MI on over 550,000 real-world ECGs, demonstrating balanced performance in a clinically imbalanced setting. The median dataset size in reviewed MI research is under 15,000 ECGs (often PTB or MIT-BIH), while our thesis leverages 550,000+ 12-lead recordings from the MIMIC-IV database, providing 35 times more data for robust model training and validation. Most reviewed studies use raw ECG waveforms only; our multimodal framework fuses raw 12-lead signals with structured clinical features (QT, PR, QRS intervals, heart rate) via attention mechanisms, enriching the feature space for improved MI detection.

In this study by Bai et al. (2024) [17], the hybrid CNN-BiGRU-Multi-Head-Attention model presented by the paper detects left bundle branch block, right bundle branch block, premature ventricular contractions and premature atrial contractions. While their five-class classifier—comprising Normal sinus rhythm, PAC, PVC, LBBB and RBBB—achieved a mean accuracy of 99.41%, precision of 99.15%, specificity of 99.68% and an F1-score of 99.21% on the MIT-BIH dataset. Our CardioFusion-RT framework, trained on over 550,000 real-world 12-lead ECGs with integrated clinical features, demonstrates superior clinical generalizability and balanced performance. Although the CBGM model reports marginally higher raw metrics on selected arrhythmia classes—such as mean accuracy and F1-scores near 99%—those results stem from evaluations on small, balanced public datasets (MIT-BIH and PTB) with limited patient diversity and artificially curated class distributions. In contrast, table 5.1 shows that our model’s slightly lower F1-scores (87–91% for the four conditions) are achieved on an order-of-magnitude larger, highly imbalanced real-world cohort of over 550,000 clinical 12-lead ECGs from MIMIC-IV, which reflects true patient heterogeneity, noise and recording variability. Consequently, our lower but robust metrics translate into stronger generalization and clinical reliability across diverse populations and rare arrhythmia presentations, whereas the CBGM’s near-perfect scores risk overfitting to controlled datasets and may not sustain performance in real-world settings.

Table 5.4: Performance Comparison of Deep Learning Models

Model	Class	Precision	Recall	F1-Score
Hybrid CNN-Transformer+FT Transformer	Normal	0.759	0.894	0.822
	Borderline	0.726	0.713	0.719
	Abnormal	0.932	0.834	0.881
	Accuracy		0.814	
CNN-BiLSTM+FT Transformer	Normal	0.776	0.909	0.837
	Borderline	0.718	0.751	0.734
	Abnormal	0.949	0.809	0.874
	Accuracy		0.819	
Densenet+FT Transformer	Normal	0.832	0.862	0.847
	Borderline	0.741	0.809	0.774
	Abnormal	0.943	0.857	0.898
	Accuracy		0.844	
CardioFusion-RT	Normal	0.850	0.845	0.848
	Borderline	0.757	0.804	0.779
	Abnormal	0.926	0.888	0.907
	Accuracy		0.851	

We implemented Resnet, Densenet, Hybrid CNN-Transformer, CNN-BiLSTM on raw ecg signals and MLP and FT-transformer was applied on the numeric features. The table 5.4 presents the performance comparison of four different deep learning architectures integrated with the FT Transformer: Hybrid CNN-Transformer, CNN-BiLSTM, Densenet and ResNet. The evaluation metrics include precision, recall, F1-score for each class (Normal, Borderline, Abnormal), along with overall accuracy. Among the four models, ResNet + FT Transformer demonstrates the most consistent and superior performance across all metrics. Specifically:

- **Accuracy:** ResNet achieves the highest overall accuracy of 0.851, outperforming Hybrid CNN-Transformer (0.814), CNN-BiLSTM (0.819) and Densenet (0.844). This indicates that ResNet generalizes better across all classes.
- **Class-wise Performance:** For the Normal class, ResNet attains an F1-score of 0.848, slightly higher than Densenet (0.847) and significantly higher than Hybrid CNN-Transformer (0.822) and CNN-BiLSTM (0.837).

For the Borderline class, which is typically the most challenging to classify due to overlapping characteristics with Normal and Abnormal ECGs, ResNet achieves the highest F1-score of 0.779. This demonstrates ResNet’s superior capability in handling difficult cases compared to the others (Hybrid CNN-Transformer: 0.719, CNN-BiLSTM: 0.734, Densenet: 0.774).

For the Abnormal class, ResNet again leads with the highest F1-score of 0.907, outperforming all other models (Hybrid CNN-Transformer: 0.881, CNN-BiLSTM: 0.874, Densenet: 0.898). This suggests ResNet’s robustness in correctly identifying abnormal conditions, which is critical in clinical applications.

- **Balanced Precision and Recall:** ResNet achieves a strong balance between precision and recall across all classes, avoiding overfitting towards a specific class. For instance, in the Abnormal class, it maintains both high precision (0.926) and recall (0.888), which ensures fewer false negatives—a crucial requirement in medical diagnostics.

Though DenseNet gives an accuracy closer to RESNET, it has some disadvantages. Densenet takes over 7 minutes to execute 1 epoch while other models need approximately 2 minutes per epoch. Densenet consumes more resources than other architectures and takes huge training time. Overall, the comparative analysis clearly shows that CardioFusion-RT consistently outperforms other state-of-the-art architectures in terms of accuracy, precision, recall and F1-score. Its superior ability to classify Normal, Borderline, and Abnormal ECG signals more effectively establishes it as the most reliable model for this study. We trained our model on both existing numeric features of machine measurement and extracted numeric features from CardioFeature Extraction Toolkit (CFET). Though we achieved slightly better accuracy with existing features, in real-world applications, it is not appropriate and suitable. That’s why we continued our work on the numeric features extracted using CFET.

5.6 Discussions

This study aimed to develop a robust deep learning framework capable of classifying ECG signals into normal, borderline and abnormal categories and further identifying specific cardiovascular diseases. The findings indicate that the proposed CardioFusion-RT architecture achieved strong performance across multiple evaluation metrics, with AUROC values exceeding 0.99 and F1-scores consistently above 0.58 for disease-specific classifications. The introduction of a three-class scheme (normal, borderline, abnormal) proved to be particularly significant. Unlike binary models, which risk misclassifying uncertain signals, the addition of a borderline class allowed the system to capture intermediate cases more reliably. This outcome directly addresses the research hypothesis that a three-class model better aligns with clinical decision-making processes and provides more actionable outputs for cardiologists. When compared to previous studies that employed binary classification, our approach demonstrated both higher diagnostic precision and better interpretability. For example, where earlier CNN-based models often reported challenges in handling imbalanced disease distributions, our integration of focal loss and threshold tuning mitigated these effects, particularly improving the detection of minority classes such as bradycardia. The combined use of raw ECG signals with numeric clinical features further strengthened classification outcomes, validating our hypothesis that a multi-modal model yields superior performance compared to signal-only methods. From a real-world perspective, these findings have important implications. First, by flagging borderline cases, the system can prompt early clinical intervention, potentially identifying at-risk patients before severe abnormalities manifest. Second, the modular nature of the disease-specific models makes the framework adaptable for clinical decision support systems in ICUs and emergency departments, where rapid ECG interpretation is critical. Third, the high specificity observed reduces the likelihood

of false alarms, which is a common limitation of automated monitoring systems.

Nevertheless, several limitations must be acknowledged. The MIMIC-IV dataset, while comprehensive, represents ICU patients and may not fully capture ECG variability in the general population. Additionally, the model’s performance is influenced by dataset imbalance, with diseases such as MI having relatively fewer samples compared to normal signals. Finally, the computational requirements of the proposed deep learning model may limit its deployment in resource-constrained settings without further optimization. Despite these limitations, the study contributes a novel perspective to ECG classification by introducing a clinically relevant three-class labeling strategy, demonstrating the effectiveness of combining CNN and Transformer-based architectures and highlighting the potential of deep learning in real-world cardiovascular diagnostics. Future work could explore expanding the model to multi-label classification of co-occurring conditions, real-time inference on wearable devices and validation on broader population datasets.

Chapter 6

Conclusion

6.1 Summary of Findings

This research successfully developed and validated CardioFusion-RT, an innovative multimodal deep learning framework that fundamentally advances cardiovascular disease classification by integrating raw 12-lead ECG signals with clinical features through a novel ResNet-FT-Transformer architecture. The framework delivered outstanding results across all evaluation metrics proving both its clinical applicability and technical excellence. The CardioFusion-RT model reached an overall classification accuracy of 85.1% with AUROC values above 0.95 for every disease category. Performance by condition was even more striking: AUROC of 0.998 for bradycardia, 0.999 for tachycardia, 0.995 for LBBB and 0.997 for RBBB. We also found that grouping results into three categories—Normal, Borderline and Abnormal helped address the uncertainty often seen in real-world ECG readings. Our dual-branch design made the most of both raw ECG time-series data and structured clinical features pulled from the CardioFeature Extraction Toolkit (CFET). The ResNet branch picked up on temporal-spatial patterns in the signals while the FT-Transformer branch captured relationships among the clinical parameters. When we compared different designs the ResNet plus FT-Transformer combo outperformed other setups including DenseNet-like models, CNN-Transformer mixes and ResNet-MLP configurations. Importantly, this final architecture balanced computational efficiency with high diagnostic accuracy making it well-suited for practical real-world use.

6.2 Contributions to the Field

This research offers several key advances in cardiovascular diagnostics and multimodal deep learning. CardioFusion-RT is the first model to successfully combine ResNet-based signal processing with FT-Transformer analysis of clinical features for ECG classification. Its dual-branch setup leverages the strengths of both convolutional and attention-based methods and fuses them effectively. Introducing a three-class system (Normal, Borderline, Abnormal) provides a more clinically meaningful alternative to traditional binary approaches by recognizing diagnostic uncertainty and offering actionable categories. The CardioFeature Extraction Toolkit (CFET) adds a transparent, reproducible pipeline for ECG feature extraction using multi-

lead beat averaging and FeatureDB delineation to ensure consistent results across studies. Our combined strategy for tackling class imbalance, weighted sampling, focal loss and threshold optimization sets a replicable standard for medical datasets with similar challenges delivering balanced performance even for rare conditions. Finally, by using mixed-precision training and optimized architectures the framework maintains computational efficiency making advanced ECG diagnostics more accessible in resource-limited settings and helping to reduce global healthcare disparities.

6.3 Recommendations for Future Work

Future work should expand CardioFusion-RT to support multi-label classification for patients with multiple heart conditions such as MI with LBBB or arrhythmia with LVH. Achieving this will involve adapting loss functions and thresholding techniques for multi-label scenarios. Creating lightweight versions of CardioFusion-RT for edge devices and wearables is another key goal using model compression methods like knowledge distillation, pruning and quantization to enable real-time ECG analysis on mobile and IoT health monitors. This would allow continuous cardiac monitoring and instant alerts for high-risk individuals. Research should also explore tracking temporal changes in ECG patterns to monitor disease progression and treatment response. Implementing federated learning would let multiple healthcare centers collaboratively improve the model without sharing patient data which addresses privacy and regulatory concerns. Additionally, integrating CardioFusion-RT outputs with clinical risk factors (age, gender, BMI, smoking status, diabetes, hypertension) and patient outcomes from MIMIC-IV could produce a cardiovascular risk scoring system. This would turn model predictions into actionable risk assessments. Finally, conducting health economic studies on deploying CardioFusion-RT evaluating savings from earlier detection, fewer diagnostic errors and better resource use will guide policy makers and support wider adoption.

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